Classifying Market Regimes

What to measure, which regimes to name, where training labels come from, and what a causal classifier can learn

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Abstract

Classifying the market regime usually means forecasting part of the future. The regimes worth building are defined by how much the market will move, which can be forecast. Regimes defined by which way it will move cannot be forecast, according to a century of US data.

A regime definition is a choice. Eight reasonable definitions of a bad market agree only moderately over a century of US data. The definitions range from bear markets to recessions to volatile months ahead. Most regime vocabularies divide five quantities into a few states: risk, the compensation for risk, drift, the economy, and co-movement. Many labels use future data. A day counts as part of a bear market because prices fell 20% afterward. From 1927 to 2025, leaving the market during bear markets known in hindsight would have raised the Sharpe ratio from 0.45 to 1.54. Leaving when the 20% rule actually confirmed each bear market gave 0.46. The confirmation came a median of 70 trading days after the peak.

Risk persists, so it can be forecast. A simple model ranked a day before a volatile month above a day before a calm month 88% of the time. Last month's volatility alone did so 84% of the time. The same model forecast the sign of the next quarter's return slightly worse than a coin flip. The VIX, credit spreads, the yield curve and the volatility term structure largely repeat what recent price variation already shows. In the worked example they added almost nothing to seven price-based features, and 19 more years of history were worth more than any extra feed. Each label source has a reach: how far into the future it looks. The reach determines how to split the training data and how late any detection will be. Official recession dates arrive 7 to 15 months late on average.

Because regimes persist, the most recent known label is a strong baseline, and a model must beat it, not a coin flip. A small, regularised model usually matches a complex one, because the data hold only a few dozen crises to learn from. A quarter-century of daily data holds nine bear markets, so evaluate per episode, with error bars that reflect the small count. The chapter recommends defining the regime by the risk a decision depends on over the next few weeks, and never training on a label before it could have been known.

What this is. The phrase "detect the market regime" contains four questions. The first is which definition of a regime the study should adopt. The second is which observable inputs carry information about it. If the study trains a model, the third is where the training labels come from. The fourth is which model turns the inputs into a classification. This chapter answers them in that order (§2–§5). Its companion, Market Regimes and Hidden Markov Models, treats a regime as a latent state and builds the hidden Markov model that infers it. This chapter treats a regime as a *target*. A decision needs the target. The study defines it explicitly and estimates it from data available at the time. Most of the chapter covers two questions that the latent-state view leaves open. These are what to feed a classifier and what to train it against.

How to read it. Part I (§1–§2) states the problem and surveys the regime definitions practitioners use. Part II (§3–§4) covers inputs and labels. A reader building a regime study will use these two sections most. Part III (§5–§8) covers classifiers, a worked example on public data from 1926 to 2025, evaluation, and the architecture of a production study. §9 is the synthesis. A reader with time for only three parts should read §1.4, §4.9 and §6. §1.4 covers the reach of a label and why detection is forecasting. §4.9 covers vetting a label before training on it. §6 covers what a causal classifier learns on public data, with labels over a century and classifiers since 1990. Appendix A defines the statistical and financial concepts that the main text uses without deriving them.

Relationship to the other notes. Market Regimes and Hidden Markov Models supplies three results used throughout. Regime models identify volatility and not expected returns. The choice between predicted, filtered and smoothed state estimates governs measured performance. The effective sample size of any regime study is its number of episodes, not its number of days. This note is the prerequisite for §5.7 and §5.8. Implied Volatility derives the VIX and the variance risk premium used in §3.3. Bonds and Bond Markets covers the credit spreads and yield curves of §3.4 and §3.5. Trend-Following in Financial Markets covers the trend estimation behind §2.4. Edge as Information relates hit rates to information. §1.4 uses that relation to size what a good direction signal delivers. Each note stands alone.

Epistemic tags. The text flags claims by status where the status changes what you should do:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement among people who have looked.
- **[Contested]** — documented, but with live disagreement about magnitude, robustness, or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. It may be right. The evidence is private or absent, and absent evidence is not evidence against it.

Untagged sentences are definitions, derivations, or arithmetic. The numbers in §6 and every number attributed to "the worked example" come from public data. `figures/rc_common.py` prints them, so you can change a parameter and rerun.

Notation. S_t is the level of a total-return price index at the close of trading day t, $r_t = S_t/S_{t-1} - 1$ its simple return and $\ell_t = \log(1 + r_t)$ its log return. $A = 252$ is the number of trading days per year. $\mathcal{F}_t$ is the information available at the close of day t. It contains data *published* by then. It excludes data that describe day t but appear later.

z_t is the vector of latent **state variables** (§1.3, §2.2). They are the quantity of risk, the price of risk, the direction of the market, the macroeconomic state and the structure of co-movement. $x_t \in \mathbb{R}^d$ is the vector of **inputs** (features) known at t. The inputs relate to the state through $x_t = \phi(z_t) + \varepsilon_t$, where ε_t is the measurement noise (§1.3). A **regime** $s_t \in \{1, \dots, K\}$ is a coarse partition of the state, $s_t = c(z_t)$, into K classes (coded $\{0, 1\}$ when $K = 2$, as for a label). Lowercase s_t is the regime and uppercase S_t the price. The two never appear in the same role. As in the companion chapter, the text sets vectors such as x_t and z_t in plain type. Σ denotes a covariance matrix.

A **label** y_t is the training target attached to day t, computed from the price path (or from other data) by a rule L: $y_t = L(S_{t-a}, \dots, S_t, \dots, S_{t+b})$. Its **lookback** is a and its **reach** is b, the number of days after t on which it depends (§1.4). For a label defined over a forward window of fixed length, $b = h$, the horizon. Labels are binary unless stated, with $y_t = 1$ marking the class of interest (the bear market, the high-volatility month). $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t)$ is the best possible probability forecast of the label, and $\widehat{p}_t$ a classifier's estimate of it. An **episode** is a maximal run of consecutive days with $y_t = 1$. A label's episode count is the number of such runs, and an episode's length is the number of days in it.

$\mathrm{RV}_t^{(n)} = \sqrt{(A/n)\sum_{i=0}^{n-1} \ell_{t-i}^2}$ is the trailing n-day realised volatility, annualised. VIX_t is the Cboe volatility index in percentage points. δ is the threshold of a turning-point rule (a 20% move unless stated). τ denotes particular dates: a peak, a trough, a confirmation, a refit. $\Lambda(u) = 1/(1 + e^{-u})$ is the logistic function. The symbol σ always denotes volatility and never a sigmoid. β is the vector of coefficients of a classifier and α the ridge penalty on them (§5.4). λ is the switching penalty of a jump model (§4.6), and κ is Cohen's kappa (§6.2). AUC, the Brier score BS and the Brier skill score BSS are defined in §7.1–§7.3 (informally in §1.4 and §5.1). In §5.11, p_{jk} with two class subscripts is a Markov transition probability, as in the companion chapter. The symbols p_t and $\widehat{p}_t$, with a time subscript, are label probabilities. A symbol used in one subsection only is defined there.

What you will be able to do. After this chapter you should be able to:

1. Turn "detect the regime" into a stated decision, a state variable, a label and a horizon, and say in advance how much of the label a classifier can learn.
2. Choose inputs for each state variable, align them to the time they were published, and transform them so that a model trained on one decade applies to the next.
3. Construct training labels from six kinds of source, and measure each label's reach, persistence, stability and episode count before training anything.
4. Build a walk-forward training set that never uses a label before it was knowable, and score a classifier against the baseline that matters.
5. Choose among threshold rules, composite indices, logistic models, tree ensembles, latent-state models and teacher–student pipelines for a given decision, and know how each one fails.

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- A. Concepts and prerequisites

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1. What classifying a regime means

1.1 The wrong intuition

Most people assume that the market is in some regime at any moment: a bull market or a bear market, risk-on or risk-off, calm or stressed. On this view the regime is a fact about the world. A good enough model with enough inputs reveals it, and the history of past regimes supplies the training labels. Under this view regime detection is fault detection: more sensors give earlier and surer detection.

The intuition fails in three ways.

1. **A regime is a definition, not a fact.** Eight reasonable definitions of a "bad" regime on the same century of US data agree only moderately with one another. Their pairwise Cohen's kappa ranges from 0.03 to 0.58 for definitions in different families (§6.2). Kappa is a chance-corrected agreement score, with 1 for perfect agreement and 0 for agreement no better than chance. No single regime history exists to recover.
2. **The definitions practitioners care about depend on the future.** In the usual sense, a day is in a bear market only if prices go on to fall 20% from the preceding peak. A classifier that must say so before that fall is complete must forecast the fall.
3. **The parts of a regime that are knowable now need measurement, not classification.** Today's volatility, drawdown and credit spread are observable. A rule computes them, so a model has nothing to learn.

One number illustrates the second point. Over 1927–2025, a strategy held the US market when the ±20% bull–bear label said "bull" and held cash otherwise. It earned a Sharpe ratio of 1.54, against 0.45 for holding the market throughout. That strategy uses the label one day ahead. Acting on the label only once the rule itself has confirmed it gave a Sharpe ratio of 0.46 (§6.3). In Sharpe-ratio terms, all of the label's value comes from the days when it cannot yet be known.

The right question is which property of the coming weeks or months matters to the decision. A second question is how well the property can be forecast from what is visible now. A third is whether a discrete regime is the best way to carry that forecast. The question "which regime are we in?" is the wrong one.

1.2 Three questions before any data

A regime study should begin with a sentence, not a dataset. The sentence answers three questions.

1. **Regime of what?** This names the asset, portfolio or strategy whose behaviour the study classifies. A regime of the S&P 500's volatility, a regime of a momentum strategy's returns and a regime of the stock–bond correlation are different objects with different inputs.
2. **For which decision?** Sizing, hedging, choosing between strategies, setting risk limits, or conditioning another model. The decision says which outcome the label must separate.
3. **At what horizon?** The decision's holding period sets the label's horizon. A daily volatility-targeting rule needs next-week or next-month risk. A strategic allocation needs the next year's macroeconomic state.

A well-posed study reads: *"Reduce equity exposure when the probability that realised volatility over the next 21 trading days falls in the top fifth of its history exceeds 0.6."* Every term in that sentence is measurable. The phrase "detect the regime" is not.

The decisions fall into three groups. The groups differ in how much a model can deliver:

Decision	Outcome the label must separate	Learnable from current data?
Risk: sizing, hedging, limits	future volatility, drawdown, correlation	largely — risk is persistent (§6.4)
Direction: timing, trend exposure	future return, bull or bear phase	barely — direction is not identified (§6.4)
Macro allocation: duration, inflation hedges, factor tilts	growth, inflation, recession	partly — slowly, with lags and revisions (§2.8)

The rest of the chapter elaborates this table. Regime studies deliver value on risk-type regimes. Most of them target direction-type regimes.

1.3 The master form

Every regime study has the same five parts, whether it is a VIX threshold or a gradient-boosted classifier trained on jump-model labels. A few hidden state variables drive the market. Observable inputs measure them with noise. A regime coarsens them into a few classes. A label approximates that coarsening using the realised price path. A classifier estimates the label from inputs available at the time.

$$
\begin{aligned}
x_t &= \phi(z_t) + \varepsilon_t && \text{(inputs observe the state, with noise)} \\
s_t &= c(z_t) && \text{(a regime is a coarse partition of the state)} \\
y_t &= L(S_{t-a}, \dots, S_t, \dots, S_{t+b}) && \text{(a label is computed from the path)} \\
\widehat{p}_t &= f(x_t, x_{t-1}, \dots) \approx \Pr\left(y_t = 1 \mid \mathcal{F}_t\right) && \text{(a classifier estimates it causally)}
\end{aligned}
$$

The state z_t is continuous and unobserved. §2.2 argues that five state variables underlie every practitioner taxonomy. They are the **quantity of risk** (volatility) and the **price of risk** (risk premia and risk appetite). They also include the **direction** of the market (its conditional mean), the **macroeconomic state** (growth, inflation and policy), and the **structure** of co-movement (correlation, dispersion and liquidity). The function ϕ says how inputs relate to these variables. The term ε_t is measurement noise. For example, the VIX measures the quantity of risk plus a time-varying premium, and a credit spread measures expected default plus the price of risk.

The regime $s_t = c(z_t)$ is a discretisation. It is useful for two reasons: it carries a discrete decision (on or off, strategy A or strategy B), and it communicates the state. It costs the information that the coarsening discards. The label y_t is the computable stand-in for the regime and is the target of the supervised model. The classifier f may use only inputs dated t or earlier.

Five parts vary between studies. They are the state variable, the coarsening c, the label rule L (including its reach b), the inputs ϕ, and the classifier f. Most named methods choose three of them and leave the other two implicit. A VIX-band study chooses the quantity of risk, a coarsening into bands, and a label with zero reach, so it needs no classifier. A bull–bear study chooses direction and a turning-point label. It usually ignores that the label's reach is months long. The comparison tables of §2.12, §4.8 and §5.12 fill in this master form.

The companion chapter's master form describes the same objects with a latent-state view. There the regime is a latent Markov state, the "label" is its posterior probability, and no external target is used for training. The two views meet in §5.7 and §5.8. A latent-state model can be the classifier, or it can produce the labels that a supervised classifier then learns.

1.4 The reach of a label, and why detection is forecasting

The **reach** b of a label is the number of days after t that the label of day t depends on. The reach alone decides what a classifier can learn, how to build the training set, and how late any detection will be.

First, suppose $b = 0$. Then y_t is a function of data available at t. The best possible forecast is the rule itself, so $p_t \in \{0, 1\}$ and the error is zero. Training a model to reproduce such a label is, at best, a slow way to compute it. "VIX above 30" and

"10% below the high" are labels of this kind. They are useful regimes, but a rule measures them and a model does not learn them.

Now suppose $b > 0$. The label depends on what happens next. The best possible forecast is the conditional probability $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t)$. Even this ideal forecaster has an irreducible error. Measured by the Brier score (the mean squared difference between forecast probability and outcome), the ideal forecaster's error is the expected conditional variance of the label:

$$\mathrm{BS}^* = \mathbb{E}\big[(p_t - y_t)^2\big] = \mathbb{E}\big[p_t\,(1 - p_t)\big].$$

The second equality is the Bernoulli variance. Given $\mathcal{F}_t$, the squared error is $(1-p_t)^2$ with probability p_t and p_t^2 otherwise, so its mean is $p_t(1 - p_t)$. Hence BS^* is zero only when $\mathcal{F}_t$ settles the label. It is largest when p_t is near one half. In expectation, no model can do better than BS^*. The size of BS^* depends on what drives the label's future part.

This yields the chapter's first organising claim. **A supervised regime "detector" forecasts its label's future part. The drivers of that future part set what the detector can learn:**

- **If the future part is volatility**, the label is highly learnable, because volatility is persistent. The log VIX has a daily autocorrelation of 0.98 since 1990, a half-life of about 35 trading days (a deviation with daily autocorrelation ρ halves in $\ln 0.5 / \ln \rho$ days). Log 21-day realised volatility on the US market since 1930, regressed on its own value a month earlier, has a slope of 0.72, a half-life of about two months. In §6.4 a forecast of the label defined by next month's volatility reaches an AUC of 0.88. (AUC is the probability that the classifier ranks a randomly chosen positive day above a randomly chosen negative one. An AUC of 0.5 is chance.)
- **If the future part is direction**, the label is barely learnable. A correlation (an information coefficient) of 0.05 between a forecast and the next period's return would be a good equity-timing signal. It corresponds to a hit rate of 51.6% (Edge as Information). In §6.4, a label defined by the sign of the next quarter's return scores an AUC below 0.5 out of sample.
- **If the future part is the macroeconomic state**, the label is learnable at low frequency from slow inputs such as the yield curve and credit spreads. Both the inputs and the labels have long publication lags and revisions.

Detection is forecasting. Reach removes the usual distinction between "detecting the current regime" and "predicting the next one". Today's data can compute any part of today's regime that they settle. Any other part is a forecast. A study that claims to detect bear markets early claims to forecast falls in prices. Judge such a study against the evidence on forecasting returns, which is discouraging (Welch & Goyal, 2008).

Random reach. Some labels have no fixed reach. A bull–bear label built from turning points assigns day t to a bear market if t lies between a peak and the following trough. The rule confirms a peak only when the price has fallen $\delta = 20\%$ below it. The label of the days after a peak is therefore settled only on the confirmation date. Its reach is the **confirmation lag**, the number of days from the turning point to its confirmation. On the US market since 1926 the median confirmation lag was 70 trading days for bear markets (interquartile range 42–146) and 41 for bull markets (§6.3). For about a fifth of all days, the label could not have been known until weeks or months later.

The stale-label baseline. Because labels persist, the most recent label that *is* fully known at t is a strong forecast of the current one. For a label with fixed reach b, that label is y_{t-b}. For a forward-volatility label with horizon h, y_{t-h} uses the returns of $(t - h, t]$. It compares trailing realised volatility with a threshold that has barely moved in h days, so it is a pure persistence forecast. For a turning-point label, the stale label is the **confirmed state**, the rule's own real-time classification. Judge a classifier by its improvement over this stale label, not over the base rate. In §6.4 trailing volatility alone reaches an AUC of 0.84 on the forward-volatility label, and the full model adds 0.04.

Two exact equivalences hold:

This	is exactly	Consequence
The real-time ±20% bull–bear rule	a hysteresis rule (separate triggers for entering and leaving a state, §5.2): a 20% drawdown from the running peak starts the bear state, a 20% rise from the running trough ends it	its "detections" are threshold crossings, made when the move is already 20% deep
The ex-post ±20% bull–bear label	the real-time rule with each switch moved back to the preceding peak or trough	the two differ only between each turning point and its confirmation: 20% of days since 1927

1.5 Dating, nowcasting and forecasting

Three tasks share the name "regime detection". They use different information, so evaluate them differently.

Task	Information used	Example	Evaluated against	Typical error
Dating	the whole sample, before and after t	NBER recession dates, the ex-post bull–bear label, smoothed HMM probabilities	historical judgement	treating dated labels as observable in real time
Nowcasting	$\mathcal{F}_t$, for a label of day t	"is today in a bear market?", this week's financial-conditions index before it is published	the label, once settled	reporting accuracy on days whose label was already settled
Forecasting	$\mathcal{F}_t$, for a label of a later day	"will next month be high-volatility?"	the realised label	ignoring overlap between consecutive labels

The three tasks correspond to the three information sets of the companion chapter (section 5 of Market Regimes and Hidden Markov Models). Dating uses smoothed information, nowcasting uses filtered information, and forecasting uses predicted information. Published regime classifiers most often make one error: they train on dated labels and report accuracy against the same dated labels. This treats the accuracy as attainable in real time. It is attainable only on days whose label had already been settled, and a classifier is least needed on those days.

Nowcasting a label with a long reach is forecasting. Nowcasting a series that arrives late is a different task. The Chicago Fed releases its National Financial Conditions Index for the week ending Friday on the following Wednesday. Estimating it on Monday from daily market data is a measurement problem with a short horizon and a known answer. Such series are good uses of a supervised model, because the target exists and arrives soon.

§1 Key takeaways

1. A regime is a definition chosen to serve a decision. Write the decision, the state variable, the label and the horizon in one sentence before touching data.
2. The reach of a label, the number of future days it depends on, sets what a classifier can learn. A label with zero reach is a rule to compute. A label with positive reach is a forecast.
3. Supervised regime detection forecasts the label's future part. Labels whose future part is volatility are learnable. Labels whose future part is direction are not.
4. The baseline that matters is the stale label: the most recent fully known label, or the rule's own confirmed state. Judge every classifier by its improvement over it.
5. The ex-post bull–bear label and its real-time version differ only between each turning point and its confirmation. That interval covers a fifth of all days, and the label's value lies in it.
6. Dating, nowcasting and forecasting are different tasks. Accuracy measured against dated labels on days already settled gives no information about real-time performance.

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2. How practitioners carve up regimes

Practitioners use a dozen regime vocabularies: volatility regimes, bull and bear markets, risk-on and risk-off, stress regimes, business-cycle phases, growth and inflation quadrants, correlation regimes. They look like different ideas. This section argues that they are coarsenings of five state variables. It describes each family with the same fields. It also shows where the families agree and where they differ. The fields answer which kinds of regime a model can predict. Families built on the quantity of risk can be forecast. Families built on direction cannot. Macroeconomic families can be forecast slowly.

2.1 A short lineage

The vocabulary accumulated over a century. Each stage added something that practitioners still use:

Year	Development	What changed
1922	W. P. Hamilton's *The Stock Market Barometer* sets out Dow theory: primary bull and bear markets, secondary reactions, and confirmation by two averages	The first explicit bull–bear taxonomy, with a confirmation lag built into the definition
1946	Burns and Mitchell, *Measuring Business Cycles* (NBER)	Expansions and contractions dated by expert judgement; the official recession dates that most macro labels still use
1971	Bry and Boschan's turning-point algorithm	Dating becomes a reproducible program; its window and minimum-duration parameters become choices
1989	J. D. Hamilton's Markov-switching model	The regime becomes a latent state with probabilities rather than a dated label
2003–04	Pagan and Sossounov; Lunde and Timmermann	Price-based bull–bear dating with explicit thresholds
2004–12	The Investment Clock; Bridgewater's All Weather	Allocation by growth–inflation quadrant
2006–19	Financial stress and conditions indices from central banks and the OFR	Composite, observable stress regimes published weekly or daily
2018–24	Jump models, clustering, and supervised classifiers trained on model labels	Regimes defined by fitted models; classifiers trained to predict them

The first row matters most for this chapter. A primary bear market in Dow theory exists only once the averages have confirmed it. So the definition that practitioners inherited lagged from the start. Ang and Timmermann (2012) review the econometric strand of this history, from Hamilton's model onward.

2.2 Five state variables under every taxonomy

The families below differ in vocabulary, but each one partitions one or two of five continuous state variables. Naming them is the first step to choosing inputs (§3), because inputs are measurements of state variables, not of regimes.

State variable	What it is	Typical observables	Persistence	Known at t?
Quantity of risk	the conditional volatility of returns	realised volatility, VIX, MOVE	high: half-life of one to two months	yes, with small error
Price of risk	compensation demanded per unit of risk; risk appetite	credit spreads, variance risk premium, valuations	high	partly: a spread is observed, its split into expected loss and premium is not
Direction	the conditional mean, or trend	trailing returns, distance from moving averages	low	no: the mean is not identified in short samples
Macroeconomic state	growth, inflation, policy stance	activity data, inflation, the yield curve, the policy rate	very high	with publication lags and revisions
Structure	how assets move together, and how easily they trade	correlations, dispersion, breadth, liquidity measures	moderate	yes, with estimation noise

The last column, together with persistence, decides what a model can predict. The quantity of risk is observable and persistent, so a regime built on it can be forecast a few weeks ahead with real skill. Direction is neither observable nor persistent,

so analysts can date a regime built on it in hindsight but can hardly forecast it. The companion chapter shows the statistical cause of this asymmetry. In a fitted regime model, the same data pin down state volatilities sixteen times more sharply than state means (section 8 of Market Regimes and Hidden Markov Models). Most practitioner regimes combine the quantity of risk with direction or with the price of risk. Their observable part is therefore largely volatility.

The following sections describe each family with the same fields. The fields are the **idea**, the **state variables**, the **definitions in use** and typical **durations and episodes**. They also include whether the regime is **observable at** t, the **decision** it serves, its **failure modes**, and **when to use it**.

2.3 Volatility regimes

Idea. Classify the market by how much it moves.

State variables. The quantity of risk.

Definitions in use. Bands on the VIX are the commonest: below 15 “calm”, 15–25 “normal”, 25–35 “elevated”, above 35 or 40 “crisis”. **[Practice]** These cut points are convention, not estimates. Since 1990 the VIX’s median close has been 17.6. It closed below 15 on 32.5% of days, above 30 on 8% and above 40 on 2.3%. Other definitions use quantiles of realised volatility (the top fifth of the history so far) or bands on a GARCH volatility estimate. Others use the calm and turbulent states of a two-state hidden Markov model fitted to returns. Others use composite scores that average standardised measures of the VIX level, its recent change and its own volatility.

Durations and episodes. Long and numerous. In the worked example, the label “next month’s realised volatility falls in the top fifth of its history so far” has 166 episodes since 1930. The median episode length is 16 days.

Observable at t? Yes for the current level. A sizing decision needs the level over the coming month, and that level is forecastable with an AUC near 0.88 (§6.4).

Decision. Position sizing and volatility targeting, option strategy selection, stop and limit widths, risk budgets.

Failure modes.

- *Non-stationary levels.* A VIX of 20 was a calm reading in 2002, when the index averaged 27. It was an alarming reading in 2017, when the index averaged
  11. Fixed bands misclassify whole years.
- *Rolling normalisation turns a level into a surprise.* A score built from a rolling z-score over, say, a year adapts to the level it measures. After a year at a VIX of 35 the score reads “normal”. Such a score classifies the *change* in risk. That is a different regime from the level of risk, and a sizing rule needs the level. For levels, use long expanding windows or percentiles against the history so far. Reserve short-window z-scores for measuring surprise explicitly (§3.12).
- *Implied is not realised.* The VIX contains a variance risk premium that varies with the price of risk (§3.3), so a VIX band mixes two state variables.
- *A volatility regime is not a direction regime.* High volatility also contains the best days of each decade, which cluster in rebounds. Volatility timing works by reducing risk. It does not work by avoiding losses. **[Contested]** Studies dispute whether it also improves returns out of sample (Moreira & Muir, 2017; Cederburg et al., 2020).

When to use it. Whenever the decision is about risk. It is the default regime, and the only family for which forecasts are reliably better than the stale label.

2.4 Trend regimes: bull, bear, correction, rebound

Idea. Classify the market by the direction and persistence of its price path.

State variables. Direction, with the quantity of risk entering because bear markets are volatile.

Definitions in use. Five constructions are common.

1. *The ±20% rule.* A bear market runs from a peak to the following trough. A fall of at least 20% must follow the peak, and a rise of at least 20% must follow the trough. The financial press uses it. Lunde and Timmermann (2004) formalised it with separate thresholds for the two directions. A fall of 10% to 20% is a *correction*.
2. *Business-cycle dating applied to prices.* Pagan and Sossounov (2003) adapted the Bry–Boschan algorithm to monthly stock prices. The method finds local maxima and minima, enforces alternation and censors the ends of the sample. It requires a minimum cycle of 16 months and a minimum phase of four months, unless the move exceeds 20%.
3. *Trailing-return states.* Cooper, Gutierrez and Hameed (2004) call the market UP when its return over the past 36 months is non-negative. Daniel and Moskowitz (2016) mark a bear state when the past two years’ cumulative return is negative.
4. *Moving-average states.* The price is above or below its 200-day or ten-month average (Faber, 2007).

5. *Four-phase cycles.* Goulding, Harvey and Mazzoleni (2023) cross the signs of a slow (12-month) and a fast (1-month) time-series momentum signal. The state is Bull when both are positive and Bear when both are negative. It is Correction when the slow signal is positive and the fast one negative, and Rebound when the slow signal is negative and the fast one positive. Maheu, McCurdy and Song (2012) fit a four-state Markov-switching model with bull, bull-correction, bear and bear-rally states.

Durations and episodes. Few. The ±20% rule finds 31 bear markets on the US total market since 1926. Nine of them occurred since 2000. The median length is about six months (128 trading days on the common sample of §6.2). §6.3 lists those since 2000. Longer histories add episodes: Gonzalez, Powell, Shi and Wilson (2005) date two centuries of US bull and bear cycles.

Observable at t? No for the ex-post definitions (1 and 2). Their reach is the confirmation lag, a median of 70 trading days for bear markets. Yes for the trailing definitions (3, 4 and 5), because they are rules with zero reach. They are trend signals, not forecasts. The trend-following literature assesses their value (Trend-Following in Financial Markets).

Decision. Market exposure. The risk of momentum strategies, whose crashes cluster in bear states with high volatility, when the market rebounds (Daniel & Moskowitz, 2016). The choice of trend speed (Goulding, Harvey & Mazzoleni, 2023).

Failure modes.

- *Thresholds and index choice flip borderline episodes.* The S&P 500 price index fell 19.9% in 1990, 19.3% in 1998, 19.4% in 2011 and 19.8% in 2018. None of these is a bear market by the press convention. The total US market including dividends and small companies fell more than 20% in all four, so each is a bear market in the worked example. Four of its eleven bear markets since 1990 exist or not depending on the index.
- *Bear-market rallies count as bull markets.* The US market rose more than 20% between late November 2008 and early January 2009 before falling to new lows. So the ±20% rule records a brief bull market in the middle of the financial crisis. Four-phase definitions exist to fix this.
- *The confirmation lag.* By construction, the rule recognises a bear market once prices are 20% below the peak.
- *Censoring.* The last phase of any sample is provisional because its end is unknown.

When to use it. For describing history and for benchmarks. Use it as a training target only if the decision needs direction and the user accepts the limit on learning direction set out in §1.4.

2.5 Trending versus mean-reverting markets

Idea. Classify the market by whether it extends moves or reverses them.

State variables. The persistence of direction: the autocorrelation of returns at the strategy's horizon.

Definitions in use. Four kinds of statistic are common. The variance ratio $\mathrm{VR}(q)$ is the variance of q-day returns divided by q times the variance of one-day returns. It is above one for trending markets and below one for mean-reverting markets (Lo & MacKinlay, 1988). The others are the Hurst exponent and the autocorrelation of daily returns. **[Practice]** Practitioner indicators include the efficiency ratio (net move divided by path length) and the average directional index.

Durations and episodes. Short and noisy.

Observable at t? Only with large error. An autocorrelation estimated from 63 days of returns has a standard error of about $1/\sqrt{63} \approx 0.13$, larger than the variation in autocorrelation that the regime should detect.

Decision. Choosing between trend-following and mean-reversion strategies, and placing stops and targets.

Failure modes. Noise dominates. The P&L of a trend rule depends on the autocorrelation of returns at its own horizon over the holding period. That autocorrelation is a forecast of direction's persistence, not a measurement of it (Trend-Following in Financial Markets).

When to use it. As a descriptive statistic over long windows. **[Practice]** Little out-of-sample evidence supports switching strategies on short-window estimates.

2.6 Risk-on and risk-off

Idea. On some days and in some periods, risky assets rise together while havens fall (risk-on). In others the reverse happens (risk-off).

State variables. The price of risk, at a global level, and structure.

Definitions in use. The sign or standardised size of a common factor in cross-asset returns. The factor is usually the first principal component of returns. The principal component is the single weighted average of the returns that carries the most

common variation. The returns are those of equities, credit, emerging markets, commodities and carry currencies, against Treasuries, gold, the yen, the Swiss franc and the dollar. Chari, Dilts Stedman and Lundblad (2023) build a risk-on/risk-off index from several market-based measures. Miranda-Agrippino and Rey (2020) find a single global factor that explains a large share of the variation in risky-asset prices worldwide. Bekaert, Hoerova and Lo Duca (2013) split the VIX into risk aversion and uncertainty.

Durations and episodes. The daily sign flips often. The strength of the common factor persists for weeks to months. That strength is the factor's share of cross-asset variance over a rolling window.

Observable at t? Yes. It describes today's co-movement.

Decision. Cross-asset hedging, the risk of carry trades and of emerging-market exposure.

Failure modes. The definition is close to circular, because a risk-off day is a day on which risky assets fell. The set of havens changes. Treasuries hedged equities from about 2000 to 2021 and did not in 2022 (§2.9). A daily sign is not a regime. Use the rolling strength of the factor, or its smoothed level.

When to use it. For multi-asset portfolios, as a structure measure rather than a forecast.

2.7 Financial stress and liquidity

Idea. Either the financial system functions normally, or funding and market liquidity are impaired and forced selling drives prices.

State variables. The price of risk and structure. In stress, correlations rise and diversification fails.

Definitions in use. Composite indices published by public institutions:

- The **St. Louis Fed Financial Stress Index** (STLFSI4) combines 18 weekly series (seven interest rates, six yield spreads and five other indicators) through their first principal component. Its average since late 1993 is zero, so positive values mean above-average stress (Kliesen & Smith, 2010).
- The **Chicago Fed National Financial Conditions Index** (NFCI) is a weighted average of 105 indicators of risk, credit and leverage, with weights from a dynamic factor model. It is scaled to mean zero and standard deviation one since 1971. Positive values mean tighter-than-average conditions (Brave & Butters, 2012). An adjusted version (ANFCI) removes the part explained by economic activity and inflation.
- The **OFR Financial Stress Index** combines 33 daily market indicators in five categories (credit, equity valuation, funding, safe assets and volatility). It is zero at normal stress. Its values do not change once estimated (Monin, 2019).
- The ECB's **CISS** weights its sub-indices by their cross-correlations. Stress therefore counts for more when it is systemic, meaning that it spreads across markets at once (Hollo, Kremer & Lo Duca, 2012).
- **Turbulence** is the squared Mahalanobis distance of today's cross-asset returns from their historical mean (§5.3; Kritzman & Li, 2010). The **absorption ratio** is the share of variance explained by the leading eigenvectors (Kritzman, Li, Page & Rigobon, 2011). Both are market-only constructions of the same kind.

[Practice] Common cut points are zero (above-average stress), and one and two standard deviations for "significant" and "severe" stress.

Durations and episodes. Weeks to months, with few systemic episodes. In the US since 1990 these were roughly 1998, 2001–02, 2007–09, 2011, 2020 and a handful of shorter ones.

Observable at t? Yes for the daily market-based indices, after a lag of a day or two. The OFR index publishes two business days late. Yes, with a longer lag, for the weekly indices, whose readings for the week ending Friday appear the following Wednesday.

Decision. Leverage, liquidity buffers, risk limits, and whether to trust a diversification assumption.

Failure modes.

- *Revisions.* The NFCI is re-estimated every week, so its history changes. The Chicago Fed describes the changes as usually small, but a backtest on today's history uses weights that were not available then.
- *Full-sample weights.* Weights estimated on the whole sample and applied to its early years carry information from later crises into earlier readings.
- *Composition changes.* The STLFSI was rebuilt as LIBOR was retired. Its LIBOR-based spreads gave way to SOFR-based ones. Version 4 replaced a backward-looking 90-day average SOFR with the forward-looking 90-day rate.
- *Conditions are not stress.* The NFCI includes leverage measures, which build slowly in good times. A stress index measures acute strain. The two answer different questions.

Evidence. The NFCI predicts episodes of financial stress at horizons up to a year (Brave & Butters, 2012). Financial conditions forecast the lower tail of GDP growth much better than its median (Adrian, Boyarchenko & Giannone, 2019).

When to use it. For risk limits and leverage, and as a compact input that summarises dozens of series (§3.6).

2.8 Macroeconomic regimes

Idea. Classify periods by the state of the economy, on the theory that asset classes respond to it in understandable ways.

State variables. The macroeconomic state: growth, inflation and policy.

Definitions in use. Four variants.

1. *The business cycle.* NBER expansions and recessions are the reference. Real-time alternatives include Markov-switching dynamic-factor models of activity (Chauvet & Piger, 2008). Another is the Sahm rule, which signals a recession when the three-month average unemployment rate rises half a percentage point above its low of the previous twelve months (Sahm, 2019). A third is probit models on the yield curve (Estrella & Mishkin, 1998). Hamilton (2011) surveys how such methods have performed in real time. Giusto and Piger (2017) date turning points in real time with a simple machine-learning classifier, vector quantisation. **[Practice]** Analysts usually read composite leading indicators such as the OECD's in four phases. The phases cross the indicator's level relative to its trend with its direction: above trend and rising, above and falling, below and falling, below and rising (OECD, 2012).
2. *Growth–inflation quadrants.* Bridgewater's All Weather framework defines four environments by whether growth and inflation are rising or falling relative to expectations. It holds assets that suit each environment, balanced by risk (Podolsky, Johnson & Jennings, 2012). Merrill Lynch's Investment Clock names four phases from the direction of growth and inflation: reflation, recovery, overheat and stagflation (Greetham and colleagues, Merrill Lynch, 2004). These are practitioner frameworks from firms that sell allocation products built on them.
3. *Statistical macro-market regimes.* A Gaussian mixture fitted to 17 factor returns from 1971 to 2020 yields four conditions named Crisis, Steady State, Inflation and Walking on Ice (Botte & Bao, 2021). Guidolin and Timmermann (2007) find four states for stocks and bonds jointly: crash, slow growth, bull and recovery.
4. *Monetary-policy regimes.* Hiking, cutting and holding cycles, measured by the six- to twelve-month change in the policy rate or the two-year yield, and the zero-rate period as its own regime.

Durations and episodes. Years, and very few. The US has had 12 recessions since 1948. Inflation regimes number a handful.

Observable at t? Only with publication lags and revisions. NBER dates are not observable in real time at all. Over the cycles since 1978, the committee announced peaks an average of 7.3 months after they occurred, and troughs 15 months after (NBER announcements). The committee announced the trough of March 1991 in December 1992.

Decision. Strategic allocation, duration and inflation hedging, factor tilts.

Failure modes.

- *Few episodes.* Twelve recessions is the whole post-war sample for a recession classifier.
- *Revisions.* Real-time and final macroeconomic data differ, sometimes enough to change the sign of a growth reading (Croushore & Stark, 2001).
- *Two-sided filters.* A detrended indicator built with the Hodrick–Prescott filter revises its recent values as new data arrive (Hamilton, 2018). The OECD's composite leading indicators have used this filter since 2008.
- *Relationships break.* The 10-year minus 3-month Treasury spread was negative from late October 2022 to mid-December 2024. This was the longest inversion on record. No NBER recession had followed at the time of writing.

When to use it. For slow allocation decisions, with labels lagged by their publication delay. Rarely for daily equity decisions, where macro data move too slowly to help except through surprises on release days (§3.10).

2.9 Correlation and dispersion regimes

Idea. Classify periods by how assets move together, not by how much they move.

State variables. Structure.

Definitions in use.

- *The stock–bond correlation.* Negative from the late 1990s to about 2020, positive before that and again from 2021–22. Its sign appears to track whether inflation shocks or growth shocks dominate (Campbell, Sunderam & Viceira, 2017). Macroeconomic fundamentals explain little of its variation, and liquidity factors explain more (Baele, Bekaert &

Inghelbrecht, 2010). For a balanced portfolio it is the most consequential regime of all (section 14 of Bonds and Bond Markets).

- *Equity correlation.* The average correlation among index members, or the implied correlation from index and single-stock options (Cboe's COR1M and COR3M indices, with history from 2006). **[Contested]** Average correlation predicts market returns, because it tracks aggregate risk (Pollet & Wilson, 2010). Correlation risk also carries a premium in index options (Driessen, Maenhout & Vilkov, 2009).
- *Dispersion.* The cross-sectional spread of stock returns separates a "macro" market from a "stock-picker's" market. It relates to time variation in the value and momentum premiums (Stivers & Sun, 2010).

Durations and episodes. The stock–bond sign changes every decade or two, so episodes are very few. Equity correlation regimes last weeks to months.

Observable at t? Yes, with estimation noise. A correlation estimated over 63 days has a standard error of roughly $(1-\rho^2)/\sqrt{63}$, about 0.11 at $\rho = 0.3$. Volatile periods also bias measured correlations upward for mechanical reasons (Forbes & Rigobon, 2002).

Decision. Diversification and hedge ratios, dispersion trades, and factor exposure.

Failure modes. Estimation noise and the volatility bias. For the stock–bond sign, the sample has two or three regimes in a century.

When to use it. Whenever a portfolio relies on a correlation. It is the regime most often ignored and most often decisive.

2.10 Calendar and event regimes

Some regimes are known in advance: Federal Reserve announcement days, the release days of inflation and employment data, monthly options expiry, quarter ends, earnings seasons, holidays. Volatility and volume behave differently on these days. Enter them in a study as features, never as something to detect. Leaving them out has a cost. A classifier that does not know a release is scheduled may read the release-day spike in volatility as the start of a regime.

2.11 How the families overlap, and where they differ

The families overlap because volatility rises when prices fall. Low volatility, bull markets, risk-on, low stress and expansions tend to occur together. The overlap is real but partial. In the worked example's century of US data, the chance-corrected agreement between the ex-post bear-market label and the next-month high-volatility label is $\kappa = 0.35$. The agreement between either of these labels and NBER recessions is about 0.3. The agreement between the rule's own real-time bear state and a negative next quarter is 0.03 (§6.2).

The episodes where the families differ are the informative ones. The table uses the worked example's data: the US total market, the VIX and the Baa–Treasury credit spread.

Episode	Largest drawdown in the window	VIX	Baa – 10-year spread, start → high	NBER recession	Families that fired
Aug–Dec 1987	33.1%	not yet published	2.0 → 2.7 points	no	volatility, trend; no macro
Jan–Jun 2007	5.9%	mean 13, high 20	flat near 1.6	no	almost none; the yield curve was inverted
Jul–Dec 2007	9.7%	mean 22, high 31	1.6 → 2.6 points	no (the peak was December)	credit and stress before equity
Sep–Dec 2018	20.4%	high 36	1.9 → 2.5 points	no	trend, volatility; mild credit
Feb–Apr 2020	34.2%	high 83	2.1 → 4.3 points	yes	all of them, briefly
Jan–Oct 2022	25.4%	mean 26, high 36.5	1.8 → 2.4 points	no	trend; stock–bond correlation turned positive
Feb–Apr 2025	19.6%	high 52	1.5 → 2.0 points	no	volatility; trend on some indices

Two lessons follow. First, a single "regime" label does not show which family fired, and the families call for different responses. The credit-led stress of late 2007 called for cutting leverage while equity volatility was still moderate. The 2022 bear market called for hedges other than Treasuries. Second, the borderline cases decide the labels. 2025 is a bear market on some indices (the Nasdaq Composite fell more than 20%) and not on the total market, which fell 19.6%. A study's decision on such cases shapes everything trained on them.

2.12 The taxonomy as a product space

Each family is a point in the space of choices in the master form of §1.3. The families do not form a hierarchy:

Family	State variable	Usual K	Usual dating	Reach of the usual label	US episodes	Decision
Volatility	quantity of risk	2–4	threshold or quantile	0 (level) or 21 days (forward)	over a hundred	sizing, risk
Bull–bear	direction	2–4	turning points	median 70 days for bear onsets	31 since 1926	exposure
Trending vs mean-reverting	direction's persistence	2	rolling statistic	0 (estimated)	noisy	strategy choice
Risk-on/off	price of risk, structure	2	factor sign or strength	0	many, short	cross-asset hedging
Stress	price of risk, structure	2–3	composite index	0, plus publication lag	about ten	leverage, limits
Business cycle	macro	2–4	expert or model	months to years	12 since 1948	allocation
Growth–inflation	macro	4	sign of changes	months	a handful	allocation
Correlation	structure	2	rolling estimate	0 (estimated)	2–3 for stock–bond	diversification

Which axis matters. The choice of state variable determines what can be forecast far more than the details of the definition. Within a family, thresholds change the episode count a great deal and the labels much less. Moving from ±20% to ±15% raises the number of bear markets from 31 to 47, yet the two labels agree with $\kappa = 0.88$. Across families, agreement falls to between 0.03 and 0.58. Spend design effort on choosing the state variable and the horizon. Tune thresholds last.

§2 Key takeaways

1. Practitioner regimes are coarsenings of five state variables: the quantity of risk, the price of risk, direction, the macroeconomic state and structure. Name the state variable before choosing inputs.
2. Regimes built on the quantity of risk are observable and persistent, so they can be forecast. Analysts can date regimes built on direction in hindsight but can hardly forecast them. Macro regimes move slowly and arrive late.
3. The ±20% bull–bear convention is index-dependent at the margin: four of the eleven bear markets on the total US market since 1990 were not bear markets on the S&P 500 price index.
4. Stress and conditions indices are good compact inputs, but the publishers revise the weekly ones, and they use full-sample weights. Prefer unrevised indices, or store the vintages.
5. Macro regimes have a dozen episodes and labels published months late. Lag them by their real announcement delay or do not use them.
6. The families overlap because volatility rises when prices fall. They differ in exactly the episodes that matter. Keep them as separate outputs rather than merging them into one label.
7. Thresholds move episode counts more than they move labels. Choose the state variable and horizon first. Tune thresholds last.

3. Inputs: observing the state variables

An input is useful to a regime classifier only through what it says about a state variable. This section covers the input families by the state variable they measure. It gives each family's mechanism, history and failure modes. It ends with two practical problems that decide whether any of it works. The first is aligning inputs to the time they became known. The second is transforming them so that a relationship estimated in one decade holds in the next. The worked example supports the section's conclusion: the input space is much smaller than the list of series suggests. Two factors carry most of the variation, and the price path itself carries most of the information.

3.1 What makes a good regime input

Six properties, in rough order of importance:

1. **Mechanism.** A reason the input moves with the state variable the label depends on. A credit spread widens with the price of default risk, and the VIX rises with expected volatility. An input with no mechanism leads to a specification search.
2. **History.** Enough history to contain many episodes of the regime. An input that starts in 2009 has no data from 2008. In the worked example, a model that gained a new input but lost nineteen years of history did worse (§6.4).
3. **Timeliness.** Published soon after the fact, at a frequency at least as high as the decision's. A monthly series cannot inform a daily decision except on release days.
4. **Point-in-time integrity.** Never revised, or revised with stored vintages. A revised series used at its latest vintage leaks future information into the past.
5. **Stationarity, or a transform that delivers it.** Convert a level that drifts across decades into something comparable over time (§3.12).
6. **Non-redundancy.** Information beyond the inputs already present. Most candidate inputs are noisy measurements of the volatility factor (§3.13).

Recommendation. Rank candidate inputs by these properties before testing any of them. Testing forty inputs against a label with thirty episodes always finds some apparent relationship. That relationship does not survive.

3.2 The price path itself

What it measures. The quantity of risk and the direction, directly.

The inputs. The inputs include realised volatility over several horizons (5, 21 and 63 days), the downside semivariance, and the drawdown from the running or one-year high. They also include trailing returns over one, three and twelve months, the distance from moving averages, realised skewness, and jump measures. With intraday data, realised variance from five-minute returns is a far more precise volatility measurement than a daily-return estimate (Andersen, Bollerslev, Diebold & Labys, 2003). The heterogeneous autoregression of daily, weekly and monthly realised variance is the standard forecasting model built on it (Corsi, 2009).

Why they come first. They are available for the longest history, they are never revised, and they measure the two state variables most labels depend on. In the worked example, seven price-path features alone reached an AUC of 0.87 for next month's volatility and 0.83 for the ex-post bear-market label. No other input set beat them by more than 0.01 (§6.4).

Failure modes. Price-path features of direction (trailing returns, distance from averages) are rules with zero reach. They describe the trend that has happened. Any forecasting value they have is the time-series momentum effect. That effect has its own literature and its own crashes (Moskowitz, Ooi & Pedersen, 2012).

3.3 The implied-volatility complex

What it measures. The quantity of risk expected by option buyers, plus the premium they pay for it.

The inputs.

- **The VIX**, the 30-day model-free implied volatility of the S&P 500, with history from 1990 (Cboe VIX methodology). Its square is the risk-neutral expectation of the next month's variance, which equals the physical expectation plus the variance risk premium (sections 5 and 6 of Implied Volatility).
- **The VIX term structure.** Cboe publishes 9-day (VIX9D, from 2011), 3-month (VIX3M, whose free history file starts in September 2009) and 6-month (VIX6M, from 2008) indices. The ratio of the VIX to the VIX3M is normally below one. A ratio above one means the curve is inverted, which marks acute stress. It was inverted on 7.7% of days from September 2009 to 2025, in 99 separate spells. The slope carries information about the returns of volatility assets, such as variance swaps and VIX futures (Johnson, 2017; Simon & Campasano, 2014).

- **The variance risk premium**: the VIX squared minus a forecast of realised variance. Analysts often proxy the forecast by trailing realised variance. **[Contested]** The premium predicts equity returns, most strongly at about a quarter's horizon (Bollerslev, Tauchen & Zhou, 2009). **[Contested]** The VIX splits into the premium and the expected variance. The premium predicts returns, and the expected variance predicts economic activity and financial instability (Bekaert & Hoerova, 2014). The disagreement concerns magnitude and stability.
- **Volatility of volatility and tails.** The VVIX (the VIX of the VIX, from 2006) and the SKEW index (the price of S&P 500 tail protection, from 1990). **[Contested]** Evidence that the SKEW anticipates crashes is weak. It measures demand for protection, which is high in calm, complacent markets as well as before falls.
- **Cross-asset implied volatility.** The MOVE index of Treasury implied volatility, crude oil (OVX), gold (GVZ), the Nasdaq-100 (VXN) and the Russell 2000 (RVX), and implied volatility on credit ETFs. These measure the quantity of risk in other markets and help when the label is about those markets.

Why it matters. The VIX is forward-looking where realised volatility is backward-looking. In the worked example the VIX alone forecast next month's volatility better than trailing realised volatility alone, with an AUC of 0.87 against 0.84 (§6.4).

Failure modes.

- *Non-stationary levels* (§2.3).
- *A premium that moves.* The VIX mixes expected volatility with the price of volatility risk. When the premium jumps, a VIX band misreads expected risk.
- *Frequent switching in term-structure ratios.* Ninety-nine inversion spells in sixteen years is a signal that switches on and off within days. Use a smoothed ratio or hysteresis.
- *Short histories.* Every term-structure input except the VIX itself starts after 2005 in free data, so a model that relies on them trains on two or three stress episodes.

3.4 Credit

What it measures. The price of default risk, and through it the price of risk in general.

The inputs.

- **Moody's Baa yield minus the 10-year Treasury yield**, daily on FRED from 1986 (series BAA10Y). Long-history monthly versions extend the comparison back much further.
- **Option-adjusted spreads of the ICE BofA high-yield and investment-grade indices.** Since April 2026 FRED publishes only a rolling three-year window of these, so you must store a long history as you fetch it, or buy it.
- **The Gilchrist–Zakrajšek credit spread and the excess bond premium**, the part of corporate spreads not explained by expected default. The premium predicts economic activity better than the spread itself (Gilchrist & Zakrajšek, 2012). The Federal Reserve publishes it monthly from 1973 (Favara, Gilchrist, Lewis & Zakrajšek, 2016).
- **Credit ETF prices and options**: high-yield and investment-grade bond funds, the ratio of a high-yield fund to a Treasury fund, and implied volatility on the high-yield fund.

Mechanism. A spread is expected default loss plus a risk premium plus a liquidity premium (sections 8 and 9 of Bonds and Bond Markets). The premium component moves with the price of risk, so credit can lead equity. In the second half of 2007 the Baa spread widened by a full percentage point while the stock market's largest fall was under 10% (§2.11).

Failure modes. *Composition drift.* Index rating mixes and durations change over decades. *Level shifts across eras.* A Baa spread of 2.5 points was a stress reading in 2018 and an ordinary one in parts of the 1980s. *Duration mismatch.* The Baa yield refers to long bonds, compared here with a 10-year Treasury.

3.5 Rates and the yield curve

What it measures. The macroeconomic state and expected policy.

The inputs. The inputs include the 10-year minus 3-month Treasury spread (daily from 1982), the 10-year minus 2-year spread and the near-term forward spread. They also include real yields and breakeven inflation rates from inflation-protected Treasuries, the 2-year yield as a proxy for expected policy, and the MOVE index.

Evidence. [Fact] The term spread is the best single predictor of recessions two to six quarters ahead for the post-war sample (Estrella & Hardouvelis, 1991; Estrella & Mishkin, 1998). A systematic comparison of leading indicators finds the term spread strongest four to six quarters ahead. Its recent path and broker-dealer margin balances add information at other horizons (Liu & Moench, 2016). The spread between the expected short rate six quarters ahead and the current short rate is a cleaner version of the signal (Engstrom & Sharpe, 2019).

For equity regimes. The curve is slow. In the worked example the 10-year minus 3-month spread was nearly uncorrelated with the volatility inputs (correlation 0.05 with the log VIX) and added nothing to daily classifiers (§3.13, §6.4).

Failure modes. The relationship rests on eight or so recessions since the 1960s. No recession has followed its latest signal so far. The inversion from October 2022 to December 2024 was the longest on record (§2.8). The slope mixes expected policy with a term premium that quantitative easing compressed. The zero lower bound held the short end of the curve near zero from 2009 to 2015.

3.6 Composite stress and conditions indices

What they measure. The price of risk and structure, summarised from dozens of series.

The inputs. The STLFSI4 and NFCI (weekly), the OFR Financial Stress Index (daily), and the ECB's CISS for the euro area (§2.7). Each is a factor model that an outside institution maintains. It compresses many spreads, rates and volatilities into one number.

Use them with three precautions. Lag each index to its publication date. The weekly indices release readings for the week ending Friday on the following Wednesday. Prefer indices that are not revised, such as the OFR index, or store the vintages of those that are, such as the NFCI (FRED's archival service, ALFRED, keeps them). Also remember that weights estimated on the full sample make the index's early history partly hindsight.

3.7 Breadth and market internals

What they measure. Structure: how many stocks take part in a move.

The inputs. The share of index members above their 50- and 200-day moving averages, the advance–decline line, and new 52-week highs minus new lows. They also include up and down volume, the NYSE TICK (stocks trading on an uptick minus those on a downtick, intraday), and the Arms index or TRIN. The last group is ratios of equal-weight to cap-weight indices, small to large companies, and cyclical to defensive sectors.

Mechanism. [Practice] Practitioners hold that narrowing participation precedes market tops and that broad "breadth thrusts" follow bottoms.

Evidence. Breadth and similar sentiment measures move strongly with contemporaneous returns but have little power to predict near-term returns (Brown & Cliff, 2004).

Failure modes.

- *Survivorship.* Breadth computed from today's index members excludes the stocks that the index removed after they fell. Historical breadth then looks healthier than it was in every downturn, and a classifier trained on it learns a false relationship. Use point-in-time membership, or treat the bias as known and avoid breadth levels.
- *Data quality.* Exchange-published breadth series have gaps and definitional changes. Derived series need consistent universes.
- *Noise.* TICK and TRIN are intraday microstructure measures. At daily horizons they are dominated by noise.

3.8 Correlation, dispersion and cross-asset structure

What they measure. Structure, and through it the price of risk.

The inputs. The inputs include the average pairwise correlation of index members, or each member's correlation with the index, over one to three months. They also include implied correlation (COR1M and COR3M), the cross-sectional dispersion of returns and the rolling stock–bond correlation. The last inputs are turbulence and the absorption ratio (§2.7) and the strength of a cross-asset risk factor (§2.6).

Evidence. Average correlation predicts market returns (Pollet & Wilson, 2010). Correlation risk is priced in index options (Driessen, Maenhout & Vilkov, 2009). **[Contested]** Turbulence and the absorption ratio rise before and during systemic episodes (Kritzman & Li, 2010; Kritzman, Li, Page & Rigobon, 2011). The authors' firm sells analytics built on them, and the evidence is largely in-sample.

Failure modes. Estimation noise (§2.9). Correlation rises mechanically with volatility, which biases the measure. A correlation regime can therefore be a volatility regime that looks like a correlation regime (Forbes & Rigobon, 2002). Survivorship in member-based measures, as with breadth.

3.9 Sentiment and positioning

What they measure. The price of risk as revealed by what investors pay for protection or by their positioning.

The inputs. Put/call volume ratios come in three forms. The *equity* ratio is dominated by trading in single-stock options. The *index* ratio is dominated by institutional hedging and is structurally near or above one. The third is the total. Other inputs are survey sentiment (the American Association of Individual Investors' weekly survey) and futures positioning reported by the CFTC. Further inputs are margin debt, fund flows, and composite indices such as that of Baker and Wurgler (2006).

Evidence. Researchers have found that extreme readings of the VIX, the put/call ratio and TRIN have contrarian predictive power for S&P 500 futures returns (Simon & Wiggins, 2001). Other work finds that past returns and volatility cause such sentiment measures, and that the measures add little once those are controlled for (Wang, Keswani & Taylor, 2006). Where option volume does predict returns, the information comes from buyers opening new positions. This volume is not public, and the public ratios do not carry it (Pan & Poteshman, 2006). **[Contested]** At best, the evidence for public sentiment measures as regime inputs is contested.

Structural breaks. The composition of option volume has changed enough that put/call ratios are no longer comparable over time. Retail option trading grew sharply from 2020 (Bryzgalova, Pavlova & Sikorskaya, 2023). Options expiring the same day grew from a small fraction of S&P 500 index option volume a decade ago to more than 60% in May 2025 (Cboe, 2025). A level of the index put/call ratio in 2025 does not mean what the same level meant in 2015.

Recommendation. If you use sentiment inputs, use them as residuals. Regress each on recent returns and volatility, and keep what those do not explain. Then test the residual against the price-path baseline. Expect little.

3.10 Macroeconomic data

What it measures. The macroeconomic state.

The inputs. The inputs include weekly initial unemployment claims, and monthly payrolls, purchasing managers' indices, industrial production and retail sales. They include composite activity indices such as the Chicago Fed National Activity Index, and the daily business-conditions index of Aruoba, Diebold and Scotti (2009). They also include inflation measures and **surprise indices**. A surprise index standardises each release's deviation from the consensus forecast and aggregates across releases (Scotti, 2016). Any economic calendar that records consensus forecasts alongside actual values supports a surprise index.

Real time. Macro data are revised, sometimes heavily, and the vintage available on a date differs from today's history (Croushore & Stark, 2001). Daily nowcasting models update an estimate of the current quarter as each release arrives (Giannone, Reichlin & Small, 2008).

For equity regimes. Macro data move the slow regimes of §2.8. For daily equity classification they matter mainly through surprises on release days.

3.11 Information timing

A daily regime study joins series that are observed at different times, published with different lags, and revised on different schedules. Most regime studies leak future information at the join. The rule is to **align every input to the time it became available, not to the date it describes.**

Input	Describes	Usable from	Revised?
Index closes, VIX family	the trading day	that day's close	no
Treasury yields and the Baa spread (FRED)	a business day	the next business day	no
ICE BofA option-adjusted spreads (FRED)	a business day	the next morning	no
NFCI, STLFSI4	the week ending Friday	the following Wednesday	NFCI weekly; check vintages
OFR Financial Stress Index	a business day	two business days later	no
Cboe put/call ratios	the trading day	that evening, after the close	no
Initial claims	the week ending Saturday	the following Thursday	yes
Payrolls, CPI	the month	the following month	payrolls twice, then annually; CPI rarely
NBER turning points	a month	4 to 21 months later	rarely
Breadth from index members	the trading day	that day	survivorship unless point-in-time

Four rules follow:

1. **Lag to publication.** Shift each series by its release lag before joining it to the price data.
2. **Store what you fetched, when you fetched it.** A store that overwrites history with the latest vintage destroys the evidence of what was known.
3. **Forward-fill weekly and monthly series from their release date**, never from their reference date.
4. **Never normalise with full-sample statistics.** A z-score that uses the full-sample mean and standard deviation carries the future into every past value.

3.12 Transforming inputs

Raw levels rarely stay comparable across decades. A model trained when the VIX averaged 27 and applied when it averages 11 extrapolates. Transforms make inputs comparable across time:

- **Logs** for volatilities and ratios of volatilities. The log VIX is closer to symmetric and mean-reverting than the VIX itself.
- **Ratios and differences** for term structures (VIX/VIX3M) and for the gap between implied and realised volatility.
- **Changes** over one to three months for slow-moving levels such as spreads and the curve.
- **Percentiles or z-scores against an expanding window** for levels whose meaning should be absolute. A VIX in the top decile of all history is high, whatever the last year was like.
- **Z-scores against a rolling window** only for inputs whose meaning is relative to recent experience. A rolling z-score measures surprise. Applied to a persistent level, it removes the level. A regime score built that way classifies changes in risk, not risk (§2.3).
- **Clipping** at four or five standard deviations. This stops one crisis from dominating the fit. It also stops a new crisis from producing an absurd extrapolation.

Every transform must be causal. It may use data up to t only. An expanding quantile threshold is causal. A full-sample quantile is not.

3.13 How many independent inputs are there?

The inputs carry fewer independent dimensions than the list suggests. Take daily data from September 2009 to 2025 and seven standardised inputs. The inputs are the log VIX, log 21-day realised volatility and the log VIX/VIX3M ratio. They also include the Baa spread, the 10-year minus 3-month spread, the drawdown from the one-year high, and the distance from the 200-day average. Their first principal component explains 54% of their joint variance. The second explains 22%. The six inputs with history back to 1990 give shares of 62% and 19%. The correlations show the structure:

	log VIX	log RV	VIX/VIX3M	Baa spread	10y–3m	drawdown	trend
log VIX	1.00						
log realised vol	0.81	1.00					
VIX/VIX3M	0.68	0.55	1.00				
Baa spread	0.32	0.29	0.12	1.00			
10y–3m spread	0.05	−0.03	0.01	0.58	1.00		
drawdown	0.74	0.75	0.57	0.22	−0.12	1.00	
trend	−0.52	−0.60	−0.48	−0.28	0.05	−0.85	1.00

The correlations form two clusters. The first is a fast **volatility and drawdown** factor (VIX, realised volatility, term structure, drawdown, trend). The second is a slow **credit and curve** factor. A twentieth series usually adds a noisier measurement of the first factor. It does not add a new dimension.

Recommendation. Build one or two composite scores per state variable. Build each from a few well-understood inputs, and give the scores to the classifier. The classifier then has five to ten inputs. Its episode count can support that many.

3.14 A data map

The table lists free sources and the depth of their daily history. The depth sets how many episodes a classifier can learn from.

Input	State variable	Free source	Daily history from	Published	Revised
US market total return	quantity of risk, direction	Kenneth French data library	1926	monthly batches	rarely
VIX	quantity of risk	Cboe	1990	same day	no
VIX9D, VIX3M, VIX6M	term structure	Cboe	2011, 2009, 2008	same day	no
VVIX, SKEW	volatility of volatility, tails	Cboe	2006, 1990	same day	no
COR1M, COR3M	implied correlation	Cboe	2006	same day	no
MOVE	rates volatility	ICE, via data vendors	1988	same day	no
Baa minus 10-year	credit	FRED	1986	next day	no
High-yield and investment-grade OAS	credit	FRED (three-year window)	rolling	next day	no
Excess bond premium	price of risk	Federal Reserve	1973 (monthly)	monthly	re-estimated
10y–3m and 10y–2y spreads	macro, policy	FRED	1982, 1976	next day	no
NFCI, ANFCI	conditions	Chicago Fed, FRED	1971 (weekly)	Wednesday	weekly
STLFSI4	stress	FRED	1993 (weekly)	Wednesday	check vintages
OFR Financial Stress Index	stress	OFR	2000	two days later	no
Put/call ratios	sentiment	Cboe	2006	same evening	no
NBER recession months	macro label	NBER, FRED	1854 (monthly)	months later	rarely

Two gaps recur. First, FRED no longer carries long, free histories of high-yield spreads. A study that needs them must store its own copy or use the Baa spread and the excess bond premium. Second, the volatility term structure, implied correlation and most cross-asset volatility indices start between 2006 and 2011. A model that uses them has seen only two or three stress episodes. Index providers launched several of those indices years after the start of their published history. Their earlier values are back-calculations from archived option prices. These values are point-in-time in their inputs, but nobody published them at the time.

§3 Key takeaways

1. Choose inputs by the state variable the label depends on. For risk labels, use volatility inputs first. For macro labels, use the curve and credit.
2. Use the price path as the first input, not the last. It has the longest history and is never revised. In the worked example, no other input set beat it by more than 0.01 of AUC.
3. The VIX is the best single forward-looking volatility input. The term structure, implied correlation and cross-asset volatility indices add little on average and have short histories.
4. Credit can lead equity in credit-led stress, as in 2007. This is the main reason to include a second factor.
5. Public sentiment measures, including put/call ratios, mostly reflect past returns. Their levels have structural breaks. Residualise them or leave them out.
6. Align every input to its publication time, store vintages, forward-fill from release dates, and never normalise with full-sample statistics.
7. Rolling z-scores measure surprise, not level. Use expanding percentiles for inputs whose level matters.
8. The input space has about two dimensions. They are a fast volatility factor and a slow credit-and-curve factor. Five to ten well-chosen inputs are enough.

4. Labels: where the training targets come from

A supervised regime classifier is only as good as the label it is trained to predict. The label receives the least thought of any part of a regime study. This section treats labels as objects with measurable properties. It starts from one decomposition. It then surveys six sources of labels with a fixed set of fields. It ends with a checklist for vetting a label. It also gives a procedure for building training and test sets that never use a label before it was knowable.

4.1 The label master form

Every label is a rule applied to a window of data around its date:

$$y_t = L(\underbrace{S_{t-a}, \dots, S_t}_{\text{past part: in } \mathcal{F}_t}, \underbrace{S_{t+1}, \dots, S_{t+b}}_{\text{future part}}).$$

The past part is observable when the decision is made. The future part is not. Three properties of every label follow from its reach b. Compute them first for a new label:

1. **The purge.** At a refit date τ, the labels of the last b days before τ are not yet known. Training on them uses information from after τ.
2. **The detection floor.** A classifier can know the future part only by forecasting it. For labels whose future part is direction, that forecast is weak, so detection lags by roughly the reach.
3. **Censoring.** The last b days of any sample have no label, or a provisional one. In a live system, these days are the present, and the labels there are always missing.

A label is a proxy for an outcome that a decision depends on. A good label meets five conditions. It separates that outcome. It persists over the decision's horizon. It changes little when its parameters change slightly. It has enough episodes to learn from. Its reach leaves something to learn. §4.9 turns these conditions into tests. these into tests.

The six sources below use the same fields. The fields are the **idea**, the **definition**, the **reach** and the **parameters**. Next are the **episodes** and **persistence** the source typically produces and its **stability**. The last fields are its **decision relevance**, its **failure modes**, and **when to use it**.

4.2 Contemporaneous rules

Idea. Label a day by a condition on data observed that day.

Definition. Examples are "the VIX closed above 30", "the market closed at least 10% below its high" and "the NFCI is above zero". Others are "the price is below its 200-day average" and "trailing 21-day realised volatility is in the top fifth of its history".

Reach. Zero. The label is in $\mathcal{F}_t$.

Parameters. A threshold, sometimes a window.

Episodes and persistence. A contemporaneous rule produces many episodes, often too many. It flickers when its input stays near the threshold. In the worked example, the rule "10% below the high" produced 132 episodes since 1930, with a median length of three days (§6.2). Hysteresis removes most of the flicker (§5.2). Hysteresis uses separate thresholds for entering and leaving.

Stability. Usually moderate. The threshold matters.

Decision relevance. Direct when the decision itself is defined by the condition, as with a risk limit that binds when the VIX exceeds 30.

Failure modes. A model trained to predict a label with zero reach learns nothing that the rule does not already compute. Two supervised uses remain legitimate. The first is *nowcasting a late-published rule*. Examples are estimating this week's NFCI from daily market data before Wednesday, and estimating the monthly Sahm indicator from weekly claims. The second is *forecasting the rule h days ahead*. This turns the rule into a forward-outcome label (§4.5).

When to use it. Use it as a regime when the decision needs only the current state. Use it also as a feature and as a baseline for everything else.

4.3 Official and expert dating

Idea. Use the turning points published by an authority.

Definition. Sources are NBER business-cycle peaks and troughs and the OECD's turning points in its composite leading indicators. Others are crisis chronologies and the bear-market dates reported by the financial press. reported by the financial press.

Reach. The reach is long and variable. It equals the announcement lag. Over the cycles since 1978, the NBER announced peaks an average of 7.3 months after they occurred and troughs 15 months after. The longest wait was 21 months, for the trough of March 1991.

Parameters. The user controls none. The committee's judgement is the parameter.

Episodes and persistence. Episodes are very few and very persistent. The US has had 12 recessions since 1948, each lasting months.

Stability. High. The NBER rarely revises dates.

Decision relevance. High when a mandate or a contract depends on the official event. Otherwise indirect.

Failure modes. The dates are set with hindsight. A classifier trained on them learns to recognise a whole recession, including its first months. At the time, those months looked like an ordinary slowdown. Used without the announcement lag, the label is a look-ahead. A backtest that conditions on it reports unattainable results (section 1.5 of Market Regimes and Hidden Markov Models).

When to use it. Use it as a benchmark. Use it as a target only for slow macro decisions, and lag it by the real announcement delay.

4.4 Turning-point rules on the price path

Idea. Date peaks and troughs from the price path itself, by rule.

Definition. The $\pm\delta$ rule of §2.4 confirms a peak when the price falls δ below its running high. It confirms a trough when the price rises δ above its running low. The label marks the days between a peak and the following trough. Variants include the Bry–Boschan and Pagan–Sossounov algorithms (§2.4). These algorithms sit within the general framework of Harding and Pagan (2002) for dating cycles from turning points. Other variants are drawdown-based phases and *directional-change* event series. A directional-change series records each reversal of a fixed size, so it measures time in price moves rather than days (Glattfelder, Dupuis & Olsen, 2011; Tsang, Tao, Serguieva & Ma, 2017). **[Practice]** Another variant is the zigzag indicator of charting software.

Reach. The reach is random. It equals the confirmation lag. For the ±20% rule on the US market since 1926, the median lag was 70 trading days for bear markets (interquartile range 42–146). For bull markets it was 41 (16–76). For ±15%, the medians were 48 and 25 days.

The rule's real-time version is a hysteresis rule on drawdown and run-up. The ex-post label is that rule's state with each switch moved back to the extreme that preceded it. The two labels differ only on the days between each turning point and its confirmation. In the worked example, these days are 20% of all days since 1927.

Two-sided smoothers have the same look-ahead in continuous form. They define trends or cycles. Examples are centred moving averages, the Hodrick–Prescott filter and band-pass filters. The smoothed value at t depends on data after t. Values near the end of the sample are revised as data arrive (Hamilton, 2018).

Parameters. The threshold δ, or the window and minimum durations of a dating algorithm.

Episodes and persistence. Episodes are few and persistent. The US market had 31 bear markets at ±20% since 1926, 47 at ±15%, and nine at ±20% since 2000.

Stability. Agreement is high, but counts are unstable. The ±20% and ±15% labels agree with $\kappa = 0.88$, but the second has 50% more episodes and shorter confirmation lags.

Decision relevance. High for exposure decisions in principle. In practice the valuable part of the label is the unknowable part (§6.3).

Failure modes. Borderline episodes flip with the index and the threshold (§2.4). Bear-market rallies become bull markets. The final phase is censored. On any date, the labels after the most recent running extreme are unknown, and training on them leaks information. A model scores well on this label only by forecasting turning points, which is forecasting direction.

When to use it. Use it to describe history and to benchmark. Use it as a training target only with the purge of §4.10 applied. Expect the classifier to mostly learn to read the drawdown. The literature compares two ways of defining bull and bear markets. It reaches a consistent split. Rule-based dating describes them better in sample. Markov-switching models forecast them better (Kole & van Dijk, 2017). Using several markets' returns jointly helps both (Liu, Maheu & Song, 2024).

4.5 Forward-outcome labels

Idea. Define the regime by what happens over a fixed window after t, chosen to match the decision.

Definition. Several forms are common.

- *Forward volatility quantile.* $y_t = 1$ if realised volatility over $(t, t+h]$ exceeds q_t, where q_t is a quantile of trailing h-day realised volatility over the history up to t. With $h = 21$ and the 80th percentile this is the "next month's volatility in the top fifth" label of the worked example.
- *Forward drawdown event.* $y_t = 1$ if the largest fall within $(t, t+h]$, measured from S_t or any later peak in the window, exceeds a level such as 10%.
- *Forward return sign or quantile.* $y_t = 1$ if $S_{t+h} < S_t$.
- *Triple barrier.* Place an upper and a lower barrier, usually scaled by current volatility, and a time limit. The label records which is hit first (López de Prado, 2018).
- *Trend scanning.* Fit linear trends over forward windows of several lengths and label by the sign of the slope with the largest t-statistic (López de Prado, 2020).

Reach. Fixed at h, or bounded by the time limit.

Parameters. The horizon and the threshold. Compute the threshold causally, with an expanding quantile rather than a quantile of the full sample. A full-sample quantile carries information about the future into every label.

Episodes and persistence. These depend on what the future part measures. In the worked example, the forward-volatility label had 166 episodes since 1930 (median length 16 days). The forward-drawdown label had 71 (median 60 days). The forward-return label had 501, with a median length of four days. A direction label flickers because direction does not persist.

Stability. Good for volatility labels. Poor for return labels.

Decision relevance. The highest of any source, because the label is the decision's outcome by construction.

Failure modes.

- *Overlap.* Consecutive labels share $h - 1$ days of outcome, so neighbouring training rows are nearly copies of each other. Standard cross-validation scatters neighbours across folds, so it measures memorisation. Two steps fix this. Purging drops the training rows whose label windows overlap the test block. An embargo of at least h days drops the rows just after the test block (López de Prado, 2018; section 11 of Market Regimes and Hidden Markov Models).
- *Full-sample thresholds*, as above.
- *Horizon mismatch.* The label has a one-month horizon, but the decision is held for a week.

When to use it. Use it whenever the decision can be written as a forward event. This chapter recommends it by default.

4.6 Labels from a fitted model

Idea. Fit an unsupervised regime model to the history, and use its state sequence as the label.

Definition. Several models can supply the label. One is the smoothed state probabilities of a hidden Markov model, computed by the backward smoother of Kim (1994) and used as soft labels. Another is the most likely state path. A third is the state sequence of a **statistical jump model**. The jump model minimises the distance of each day's features from its state's centroid. It also adds a penalty λ for every switch (Bemporad, Breschi, Piga & Boyd, 2018; Nystrup, Lindström & Madsen, 2020; Aydınhan, Kolm, Mulvey & Shu, 2024). A fourth is clusters from k-means or a Gaussian mixture (Botte & Bao, 2021), including clustering of whole return distributions with the Wasserstein distance (Horvath, Issa & Muguruza, 2024). The last is the segments of a change-point method.

Reach. Formally the reach is the whole sample, because a two-sided fit lets every observation influence every label. In practice, the data within a few persistence lengths of t settle the label of day t. The worked example measures the reach for a two-state jump model. The model uses exponentially weighted features with half-lives of 10 and 42 days and $\lambda = 40$. The model's forward-only state path uses no data after t. It agreed with the two-sided labels on 94% of days ($\kappa = 0.84$). It recognised entry into the high-risk state a median of eight trading days late (interquartile range 6–11). That is a reach of about a week and a half, against months for the ±20% rule.

The parameters leak a second way. A model fitted to the whole sample defines its states using data from after t, even when its state assignment is causal. Refit the labelling model on data up to each refit date.

Parameters. The number of states, the features, and the persistence penalty or transition prior. The penalty sets the episode count directly. Setting λ to 10, 40 and 100 gave 112, 54 and 29 high-risk episodes on the century of data. Their median lengths were 30, 68 and 132 trading days.

Episodes and persistence. The penalty sets them.

Stability. Moderate. Labels near the end of a fitting window are revised as data arrive, and states can swap identities between refits. Identify each state by a fixed property, such as the lower mean return or the higher volatility.

Decision relevance. Indirect. The model defines the regime. A likelihood or a clustering objective usually makes that definition volatility. That suits risk decisions and not much else.

Failure modes. The two leaks above. The fitting objective chooses the definition instead of the decision. Persistent labels also make a classifier's accuracy look high, because yesterday's label predicts today's.

When to use it. Use it when no decision-defined label exists and a persistent summary of many features is wanted. Use it also as the teacher in a teacher–student pipeline (§5.8).

4.7 Strategy-outcome labels

Idea. Label a day by which of several actions would have done best after it.

Definition. The label can record which of a set of strategies earned the highest risk-adjusted return over $(t, t + h]$. It can record whether a base strategy's trade opened at t made money, which is the **meta-label** of López de Prado (2018). It can also record whether a risky asset beat a haven over the next month.

Reach. h.

Parameters. The set of strategies, the horizon, and the scoring rule.

Episodes and persistence. As noisy as the return differences they compare.

Stability. Low when strategies' returns are close.

Decision relevance. Maximal for strategy selection, because the label is the decision.

Failure modes. A strategy-outcome label is a direction label, because it depends on future returns. Choosing among many strategies inflates the apparent skill of the selector. Make the comparison net of costs.

When to use it. Use it when the regime study exists to switch between strategies. The meta-label is the better-behaved form. It asks only whether to act on a signal you already have. That question is narrower than which strategy is best.

4.8 Comparison of label sources

The table shows the fields that distinguish the sources. The learnability ratings are this chapter's assessment, informed by the worked example.

Source	Reach	What a classifier can learn	Typical US episodes	Persistence	Main failure
Contemporaneous rule	0	nothing: compute it	many	low without hysteresis	flicker
Official dating	months to years	slow macro forecasts	12 recessions since 1948	very high	used without the announcement lag
Turning-point rule	random, median 70 days at bear onsets	mostly the current drawdown	31 bears since 1926	high	the value lies in the unknowable days
Forward outcome, volatility	h	a great deal (AUC near 0.88)	over a hundred	moderate	overlap ignored in validation
Forward outcome, drawdown	h	some (AUC near 0.65)	dozens	moderate	overlap; few severe events
Forward outcome, return	h	almost nothing (AUC near 0.45)	hundreds of flickers	very low	mistaken for a regime
Fitted model	whole sample, about eight days in practice	the model's volatility state	set by the penalty	high	two-sided labels, full-sample parameters
Strategy outcome	h	as much as the return gap allows	depends	low	a direction label in disguise

4.9 Vetting a label before training anything

Five checks follow, each with a number to compute. A better model cannot rescue a label that fails one of them.

1. **Decision relevance.** Compare the outcome the decision depends on across the label's classes. Examples are the strategy's forward Sharpe ratio, the forward volatility and the forward drawdown. If the classes do not separate the outcome, the label is wrong for this decision. A volatility label that separates forward volatility but not the strategy's returns serves a sizing decision, not a timing one.
2. **Episodes and persistence.** Count the episodes and their median length. The median length should exceed the decision's holding period. The training set should have at least a few dozen episodes. **[Practice]** Fewer than about twenty training episodes leave too little to learn from. The test period needs its own episodes as well (§4.10).
3. **Stability.** Recompute the label with its parameters moved by a fifth to a half. For example, move the threshold from 20% to 15%, or the horizon from 21 to 15 or 30 days. Measure Cohen's kappa against the original. **[Practice]** A kappa below about 0.7 means the parameters determine most of the label.
4. **Reach and censoring.** Compute the reach, either fixed or as the distribution of confirmation lags. The reach sets the purge, the embargo and the expected detection lag.
5. **The value of knowing it, and when.** Compute what the decision would earn with the label known one day ahead (the oracle). Compute it also with the label known when it becomes known (the stale label). The oracle sets the upper bound. The stale label sets the lower bound that any classifier must beat. The gap between them is the value a classifier can add. For the ±20% bull–bear label, the upper bound was a Sharpe ratio of 1.54 and the lower bound 0.46, against 0.45 for doing nothing (§6.3). A classifier that recovers a fifth of that gap would be remarkable. The usual result is a classifier that recovers none.

4.10 Building the training and test sets

The procedure below prevents the leaks specific to regime labels. Each step names the leak it closes.

1. **Point-in-time features.** Align every input to its publication time and use stored vintages for revised series (§3.11). *Closes: look-ahead in the inputs.*
2. **Causal label parameters.** Compute thresholds with expanding windows. The outcome of a forward label lies in the future by design. Its threshold must not. *Closes: look-ahead in the definition.*
3. **Train only on labels final at the refit date.** At refit date τ, include day t only if its label was settled by τ. For a fixed reach, require $t \leq \tau - b$. For a turning-point label, allow t up to the current running extreme (the latest high in a bull

phase, the latest low in a bear phase). For a model-derived label, refit the labelling model on data up to τ and use only labels older than its effective reach. *Closes: the purge.*

4. **Walk forward.** Refit on an expanding or rolling window at a fixed cadence (quarterly or yearly) and predict the next period. When data are too scarce for that, use purged k-fold cross-validation with an embargo of at least the reach. *Closes: evaluation on the training period.*
5. **Keep episodes whole.** Never let one episode contribute rows to both the training and the test side of a split. With overlapping labels, a split in the middle of an episode puts nearly identical rows on both sides. *Closes: memorisation scored as skill.*
6. **Weight for overlap.** Down-weight rows whose labels share outcomes, using the average uniqueness of each label (López de Prado, 2018), or keep every b-th row. This changes the point estimates of a well-specified model little. Without it, the effective sample size is overstated by a factor of b.
7. **Leave class balance alone if you need probabilities.** Oversampling the rare class distorts predicted probabilities. Use class weights only if the decision uses ranks rather than probabilities, and recalibrate afterwards.
8. **Size the test set by episodes.** A test period should contain at least ten episodes of the minority class. Report results per episode as well as per day.
9. **Prefer soft labels where they exist.** Smoothed state probabilities carry the labelling model's own uncertainty. Training against them wastes less information than training against their rounded values. This is knowledge distillation (Hinton, Vinyals & Dean, 2015).
10. **Know where the label noise is.** Classifiers tolerate label noise that is unrelated to the features (Frénay & Verleysen, 2014). Regime label noise is related to the features. It concentrates at transitions, exactly where the features are changing. Expect a classifier to be least reliable at the moments that matter most.

§4 Key takeaways

1. Every label is a rule on a window of data around its date. Compute its reach first. The reach sets the purge, the detection floor and the censoring.
2. Contemporaneous rules have zero reach. Compute them. Train a model on them only to nowcast a late-published value or to forecast the rule ahead.
3. Official and expert dates are set with hindsight. Lag them by their real announcement delay or do not train on them.
4. Turning-point labels are a real-time hysteresis rule with its switches back-dated. The back-dated days are a fifth of history and hold all of the label's value.
5. Forward-outcome labels are the default, because each is the decision's outcome by construction. Their thresholds must be causal. Their overlap must be purged.
6. Model-derived labels inherit two leaks: two-sided state assignment and full-sample parameters. Refit the labeller at every refit date.
7. Vet a label before modelling it: decision relevance, episodes, stability, reach, and the gap between the oracle and the stale label.
8. Train only on labels that were final at the refit date, keep episodes whole across splits, and size test sets by episodes.

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5. Classifiers

After choosing inputs and vetting a label, the remaining choice is the model. The model matters less than the two earlier choices. In the worked example, going from a one-input to a twelve-input model added 0.04 of AUC on the volatility label (0.84 to 0.88). Moving from the volatility label to a direction label cost 0.44 (0.88 to 0.44), eleven times as much (§6.4). This section surveys the models with a fixed set of fields. The fields are the **idea**, the **definition**, the **assumptions** (and which are false in practice), **strengths**, **weaknesses**, **cost**, **robustness**, **failure modes** and **when to reach for it**. The section begins with the baselines, because a classifier is only as good as its margin over them. It ends with the post-processing that turns a flickering probability into a usable regime.

5.1 The baselines every classifier must beat

Two baselines matter. Score a classifier against both.

Climatology predicts the base rate of the label in the training data, the same probability every day. Its skill is zero by definition. The Brier skill score (§7.1) measures improvement over it: $\text{BSS} = 1 - \text{BS}_{\text{model}}/\text{BS}_{\text{climatology}}$.

The stale label predicts the current label from the most recent label already known (§1.4). For a forward-volatility label, it is trailing realised volatility. For a turning-point label, it is the confirmed state. It captures persistence, and persistence is most of what any regime classifier exploits. In the worked example, the stale label for next month's volatility already had a Brier skill score of 0.34 and an AUC of 0.84 (§6.4).

Recommendation. Report skill over climatology for comparability with the literature, and skill over the stale label for the decision. A model that does not beat the stale label only reproduces it, more slowly.

5.2 Threshold rules with hysteresis

Idea. Switch regime when an observable crosses a threshold, with a gap between the entry and exit thresholds so the regime does not flicker.

Definition. For a single input x_t and thresholds $u_{\text{out}} < u_{\text{in}}$:

$$s_t = \begin{cases} 1 & \text{if } x_t \geq u_{\text{in}}, \\ 0 & \text{if } x_t < u_{\text{out}}, \\ s_{t-1} & \text{otherwise.} \end{cases}$$

The rule enters the regime at u_{in} and stays in it until the input falls below u_{out}. With $u_{\text{out}} = u_{\text{in}}$, the rule is a plain threshold and flickers. The real-time ±20% bull–bear rule is this rule with a watched input that changes with the state. The drawdown from the running peak starts the bear state at 20%. The run-up from the running trough ends it at 20%.

Assumptions. The regime is a monotone function of one observable. This holds for volatility regimes and fails for most others.

Strengths. Transparent, with no estimation and no training labels. Easy to monitor. Hard to overfit.

Weaknesses. It uses one input. Its fixed thresholds misclassify when the input's level drifts (§2.3).

Cost. Negligible.

Robustness. High to noise, given hysteresis. Low to level drift.

Failure modes. Thresholds set on the full sample leak. Thresholds that are never revisited misclassify whole decades. Without hysteresis, the rule flickers.

When to reach for it. Use it when the regime *is* the observable, as with "VIX above 30". Use it also as the baseline for every other model.

5.3 Composite indices

Idea. Combine several standardised inputs into one score, then threshold the score.

Definition. Let $\tilde{x}_{j,t}$ be the causally standardised j-th input, signed so that a higher value means more of the state variable (more risk, more stress). The index is

$$I_t = \sum_{j=1}^{d} w_j \, \tilde{x}_{j,t},$$

The weights w_j are equal, or come from the first principal component, from a dynamic factor model (the NFCI), or from cross-correlations (the CISS). Turbulence is a related quadratic measure that uses the inverse covariance matrix. Let R_t be the vector of returns on m assets. Estimate their mean $\bar{R}$ and covariance Σ from data up to t. Then $D_t = (R_t - \bar{R})^\top \Sigma^{-1} (R_t - \bar{R})$. This is the squared Mahalanobis distance of today's returns from their mean. For one asset, it is the squared z-score. In general, it is large on days when assets move unusually far *or* in unusual combinations.

Assumptions. The inputs share one dominant factor, and that factor is the state variable of interest. For volatility and stress this holds well (§3.13).

Strengths. It uses many inputs with few parameters. It averages away input-specific noise. It needs no labels.

Weaknesses. It summarises co-movement among the inputs. It does not summarise the outcome a decision depends on. So check its relationship to the label.

Cost. Low.

Robustness. [Practice] Equal weights are more robust than estimated weights when episodes are few.

Failure modes. Three failures occur. Weights or normalisations are estimated on the full sample. An input changes its behaviour (the put/call ratio, §3.9) and changes the index without notice. Rolling standardisation turns the index into a surprise measure.

When to reach for it. Use it to build the state scores of §3.13 that a classifier then uses. Use it as the regime itself when no label is available.

5.4 Logistic and probit models

Idea. Model the probability of the label as a smooth, monotone function of a weighted sum of inputs.

Definition. For a binary label, the model is

$$\Pr(y_t = 1 \mid x_t) = \Lambda(\beta_0 + \beta^\top x_t), \qquad \Lambda(u) = \frac{1}{1 + e^{-u}},$$

The fit uses maximum likelihood with a ridge penalty. On standardised inputs, the coefficients minimise the negative log-likelihood summed over the n training rows plus $\frac{1}{2}\,\alpha\, n \|\beta\|^2$. The model does not penalise the intercept β_0. Dividing by n, the objective is the mean negative log-likelihood plus $\frac{1}{2}\,\alpha\|\beta\|^2$. So α means the same at any sample size. The penalty shrinks the coefficients towards zero. At $\alpha = 0$ the fit is plain maximum likelihood. As α grows, the model converges to climatology ($\beta = 0$, the base rate).

The model has three variants. The probit model replaces Λ with the normal distribution function and behaves almost identically. For ordered regimes (low, normal, high, crisis), the ordered logit uses one coefficient vector and $K - 1$ increasing cut points γ_k. It gives $\Pr(s_t \le k \mid x_t) = \Lambda(\gamma_k - \beta^\top x_t)$. **Dynamic** binary models add the lagged label or a lagged latent index to the right-hand side. They are the standard tools for recession and bear-market forecasting (Kauppi & Saikkonen, 2008; Nyberg, 2013; Chen, 2009).

Assumptions. The log-odds of the label are linear in the inputs. The assumption is false at the extremes. So the inputs are logged, clipped and standardised first. standardised first.

Strengths. It gives calibrated probabilities when the model is roughly right. It has few parameters, so it is stable with few episodes. A human can read its coefficients.

Weaknesses. It has no interactions unless added by hand. It is monotone in each input.

Cost. Negligible.

Robustness. Good with shrinkage, poor without it. In the worked example, a penalty of $\alpha = 0.001$ per training row gave the twelve-input model an AUC of 0.51 on the drawdown label. A one-input model fitted the same way scored 0.66. With $\alpha = 0.1$ the twelve-input model scored 0.64. Shrinkage matters because the effective sample is the number of episodes, not the number of rows.

Failure modes. A dynamic model fails when it uses a lagged label without lagging it by its reach and publication delay. A lagged NBER indicator is not known at $t - 1$. A linear model also fails at extrapolation. An input outside its training range produces extreme probabilities.

When to reach for it. Use it by default. **Recommendation:** start every supervised regime study with a regularised logistic model on five to ten transformed inputs. Require any more complex model to beat it out of sample.

5.5 Tree ensembles

Idea. Combine many decision trees, each splitting the input space at thresholds, to capture non-linear effects and interactions.

Definition. Random forests average deep trees fitted to bootstrap samples. Gradient boosting fits shallow trees in sequence, each to the errors of the ensemble so far.

Assumptions. Enough examples near each split to estimate it. With regime labels, “examples” means episodes.

Strengths. Trees find interactions such as "high volatility *and* a downtrend" without specifying them. They are insensitive to monotone transforms of the inputs. They tolerate uninformative inputs (Grinsztajn, Oyallon & Varoquaux, 2022).

Weaknesses. Trees have more parameters to fit from the same few episodes. Their predicted probabilities need recalibration. They do not extrapolate. A tree treats a VIX of 80 like the highest VIX in its training data.

Cost. Moderate. Trivial at daily frequency.

Robustness. Sensitive to the depth, the learning rate and the number of trees. Tune them with purged validation.

Failure modes. Trees overfit to a handful of crises that dominate the high-volatility region of the input space. Their stepwise response flips the regime at an arbitrary split.

When to reach for it. Use it for labels with a hundred or more episodes, such as volatility labels, and for heterogeneous inputs. The best-known teacher–student study uses it as its classifier (§5.8).

5.6 Sequence models

Idea. Let a recurrent or attention-based network learn its own features from the raw history of the inputs.

Definition. Recurrent networks, temporal convolutional networks and transformers mapping a window of past inputs to label probabilities.

Assumptions. Enough data to learn features that hand-made ones miss.

Strengths. No feature engineering. Can find temporal patterns.

Weaknesses. They need much data, and regime labels offer only a few dozen episodes. They are opaque. They are unstable across training runs.

Cost. High.

Robustness. Low with small samples.

Failure modes. They memorise the few crises in the training data. Normalisation layers fitted on the full sample leak information.

When to reach for it. Use it rarely for regime labels. The most convincing use of learned macroeconomic states is inside a larger asset-pricing objective, not as a standalone regime classifier (Chen, Pelger & Zhu, 2024). **[Practice]** A network that fails to beat a regularised logistic model with good inputs is the expected outcome.

5.7 Latent-state models with observable drivers

Idea. Skip the labels. Let a hidden Markov model or Markov-switching model infer the regime from the inputs, with persistence built in.

Definition. Candidates are a hidden Markov model whose emissions are the multivariate inputs and a Markov-switching model whose transition probabilities depend on observables (Diebold, Lee & Weinbach, 1994; Filardo, 1994). A third candidate is a Markov-switching dynamic factor model of economic activity (Chauvet & Piger, 2008). The companion chapter covers estimation in detail (section 5 of Market Regimes and Hidden Markov Models).

Assumptions. A small number of states with constant emission distributions and Markov persistence.

Strengths. The model provides probabilities and persistence. It needs no labels. It gives a predictive density.

Weaknesses. The fitted states take whatever meaning the likelihood favours, usually volatility. The model estimates transition probabilities from the few transitions in the sample.

Cost. Moderate.

Robustness. Heavy-tailed emissions and persistence priors improve it (section 6 of Market Regimes and Hidden Markov Models).

Failure modes. Smoothed probabilities fail when used as real-time signals. States can change meaning between refits.

When to reach for it. Use it when no external label suits the decision and a probabilistic, persistent state is wanted. Its filter is also the post-processor of §5.11, where it smooths another model's output.

5.8 Teacher–student pipelines

Idea. Use an unsupervised model to define the regimes (the teacher), then train a supervised classifier on causal inputs to predict them (the student).

Definition. Fit a statistical jump model or hidden Markov model to features of past returns. Its two-sided state sequence becomes the label. Train a classifier, often gradient boosting, on inputs known at t, including cross-asset and macroeconomic data the teacher never saw. Shu, Yu and Mulvey (2024b) label bullish and bearish regimes for each of twelve assets with jump models. They forecast the regimes with gradient-boosted trees using asset-specific and cross-asset macro features. **[Contested]** They report better portfolios across several allocation methods from 1991 to 2023. This is one study, by the method's developers, and it is at odds with the small gains from cross-asset inputs in the worked example (§6.4). The same group uses jump-model regimes for downside-risk reduction (Shu, Yu & Mulvey, 2024a) and for factor allocation (Shu & Mulvey, 2024). Earlier studies used the same design with other teachers. Hauptmann, Hoppenkamps, Min, Ramsauer and Zagst (2014) forecast the filtered probabilities of a two-regime Markov-switching model of monthly S&P 500 returns with logistic regressions. Uysal and Mulvey (2021) estimate the probability of a recession or a stock-market contraction with random forests on a large macroeconomic database, out of sample from 1973.

Assumptions. The teacher's regimes matter for the decision, and the student's inputs carry information about the teacher's labels beyond what the teacher's own causal state already has.

Strengths. The labels are persistent and need no external definition. All supervised methods become available. The student can use inputs the teacher could not model.

Weaknesses. The pipeline inherits two leaks from model-derived labels (§4.6): two-sided state assignment and full-sample parameters. The student learns to predict the teacher, not the outcome. If the teacher's states are volatility states, the student is a volatility forecaster with a more complicated pipeline.

Cost. Moderate.

Robustness. Depends on the teacher's stability across refits.

Failure modes. Scoring a student against labels from a teacher fitted on the whole sample overstates its accuracy near transitions. The next refit also revises labels near each refit date.

When to reach for it. Use it when a persistent regime definition is wanted and a source of labels is needed. Follow this procedure. Refit the teacher at each refit date on data up to that date. Train the student only on labels older than the teacher's effective reach. Score the student against the teacher's own causal state as the stale-label baseline. The method is a form of knowledge distillation (Hinton, Vinyals & Dean, 2015). Training against the teacher's soft probabilities rather than its hard states keeps the teacher's uncertainty.

5.9 Change-point alarms

Idea. Detect that the data-generating process has changed, without saying which of K known regimes it changed to.

Definition. Cumulative-sum (CUSUM) detectors accumulate evidence of a shift in mean or variance and alarm at a threshold. Bayesian online change-point detection maintains a posterior over the run length, the time since the last change. The posterior updates with each observation (Adams & MacKay, 2007).

Assumptions. Segments are internally stable and changes are abrupt.

Strengths. It answers "did something just change?", which is often the operative question. It controls false alarms explicitly. It needs no labels.

Weaknesses. It does not identify the new state. It is non-recurrent by design.

Cost. Low to moderate.

Robustness. Sensitive to the assumed size of the change and to the prior on run length.

Failure modes. It alarms on every volatility burst in fat-tailed data unless the observation model has heavy tails.

When to reach for it. Use it as a monitoring layer beside a classifier. An alarm signals that the inputs' behaviour has shifted and prompts a review or a reduction in size.

5.10 Skipping the regime

Idea. Forecast the quantity the decision needs directly, and discretise only for communication, if at all.

Definition. For a sizing decision, regress next month's volatility on trailing realised volatility at daily, weekly and monthly scales (Corsi, 2009) and on the VIX. Size inversely to the forecast. For a drawdown decision, forecast a low quantile of next quarter's return with quantile regression on stress inputs. Growth-at-risk does this for GDP (Adrian, Boyarchenko & Giannone, 2019).

Assumptions. The decision is a smooth function of the forecast.

Strengths. Coarsening discards no information. No labels need construction. The forecast can be evaluated directly.

Weaknesses. It is less interpretable than a named regime. Discrete decisions still need a threshold somewhere.

Cost. Low.

Robustness. High.

Failure modes. The same input-timing leaks as everything else.

When to reach for it. Use it first. **Recommendation:** build the continuous forecast before the regime classifier. Adopt a discrete regime only where it improves the decision over the forecast it coarsens. The companion chapter reaches the same conclusion from the latent-state side. It says to size continuously and not to gate (section 12 of Market Regimes and Hidden Markov Models).

5.11 Post-processing: calibration, smoothing and hysteresis

A classifier's daily probability is noisy, and a regime should not change every time the probability crosses one half. Four post-processing steps follow, in increasing order of structure:

1. **Calibration.** Map raw scores to probabilities that match observed frequencies. Fit a logistic curve to the scores (Platt scaling) or a monotone step function (isotonic regression). Fit it on training-period data only. For a flexible model, fit it on out-of-fold scores, because such a model's scores on its own training rows are too confident. Tree ensembles need calibration most (Niculescu-Mizil & Caruana, 2005).
2. **Smoothing.** An exponentially weighted average of the probability trades a few days of delay for stability.
3. **Hysteresis.** Enter the regime when the probability exceeds, say, 0.6, and leave when it falls below 0.4 (§5.2).
4. **A Markov filter over the classifier.** Treat the classifier's output as evidence and run it through a persistent Markov chain. Let π_k be the share of class k in the classifier's training data. Let p_{jk} be the probability of moving from class j to class k in a day. The filtered belief $\tilde{\xi}_t$ then updates as

$$\tilde{\xi}_t^{(k)} \;\propto\; \Big(\sum_j p_{jk}\, \tilde{\xi}_{t-1}^{(j)} \Big) \frac{\hat{p}_t^{(k)}}{\pi_k}.$$

   The ratio $\hat{p}_t^{(k)} / \pi_k$ is the classifier's evidence for class k, with its training prior divided out. The sum carries yesterday's belief forward through the chain. This is the Hamilton filter (section 5 of Market Regimes and Hidden Markov Models) with the classifier in place of the emission density. Speech recognition combined neural networks with hidden Markov models in the same way (Bourlard & Morgan, 1994). The stickiness of p_{jk} sets the trade between delay and flicker. With no persistence ($p_{jk} = \pi_k$ for every j), the filter returns the classifier's own probability. The stickier the chain, the more consecutive evidence it takes to move the belief. The filter treats each day's evidence as independent given the state. Features built from overlapping windows are not independent, so read the filter's output as a smoothed score rather than a calibrated probability.

For more than two ordered classes, smooth the cumulative probabilities rather than each class separately, so that the classes remain ordered.

5.12 Comparison

The ratings are this chapter's assessment.

Method	Needs labels	Output	Persistence	Episodes needed	Main failure	Reach for it when
Threshold with hysteresis	no	state	via hysteresis	none	level drift	the regime is the observable
Composite index	no	score	via smoothing	none	full-sample weights	building state scores
Logistic or probit	yes	probability	via inputs or a filter	dozens	extrapolation; unpurged lags	the default supervised model
Tree ensemble	yes	score	via a filter	a hundred or more	few episodes per split	many episodes, many inputs
Sequence network	yes	probability	learned	many hundreds	memorising crises	seldom
Latent-state model	no	probability	Markov	dozens of transitions	states mean volatility	no external label fits
Teacher–student	from a model	probability	from the teacher	as the student needs	two leaks from the teacher	a persistent definition is wanted
Change-point alarm	no	run length	by construction	none	fat-tail false alarms	monitoring
Direct forecast	no	quantity	via inputs	none	input leaks	first, always

§5 Key takeaways

1. Judge every classifier against the stale label as well as against the base rate. Persistence is most of what any regime classifier exploits.
2. Start with a regularised logistic model on five to ten transformed inputs. Shrinkage matters because the effective sample is the number of episodes.
3. Tree ensembles justify their complexity only on labels with a hundred or more episodes. Sequence networks rarely do.
4. Use composite indices to handle many inputs. Build a few state scores and feed them to a small classifier.
5. Teacher–student pipelines inherit their teacher's leaks. Refit the teacher at each refit date and score the student against the teacher's causal state.
6. Forecast the decision variable directly before building a regime, and keep the regime only if it improves the decision.
7. Post-process probabilities: calibrate, then smooth with hysteresis or a Markov filter whose stickiness sets the trade between delay and flicker.

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6. A worked example on a century of data

This section tests the chapter's argument on public data. It builds eight definitions of a "bad" regime on a century of US stock returns. It measures how much they agree. It measures how much hindsight is worth in the most popular one. It then trains causal classifiers with and without cross-asset inputs, to measure what each kind of label allows them to learn. The script `figures/rc_common.py` prints every number. The scripts `figures/rc_labels.py` and `figures/rc_classifier.py` draw the figures.

6.1 Data and protocol

Data. The price path is the daily total return of the value-weighted US stock market from the Kenneth French data library, July 1926 to December 2025, 26,151 trading days. Until 1952 the library includes Saturday sessions, so the early years have up to about 300 trading days a year, and a 252-day "year" is shorter there. The VIX (from 1990) and the VIX3M (from September 2009) come from Cboe's public history files. The Baa–Treasury spread (BAA10Y, from 1986), the 10-year minus

3-month Treasury spread (T10Y3M, from 1982) and the NBER recession indicator (USREC) come from FRED. The analysis lags the two FRED spreads by one day, to match their publication time.

Labels. The study uses nine labels, each marking a "bad" class. They are eight definitions and the ±15% variant of the first.

1. *Bear, ±20% rule, ex post*: days from a peak to the following trough. Moves of 20% confirm peaks and troughs (§4.4).
2. *Bear, ±20% rule, as confirmed*: the same rule's real-time state.
3. *Bear, ±15% rule, ex post*: as the first, with 15% thresholds.
4. *10% below the high, now*: the market closed at least 10% below its all-time high.
5. *Next-month volatility, top fifth*: realised volatility over the next 21 trading days at or above the 80th percentile of trailing 21-day realised volatility up to t. The threshold expands, and it applies after a burn-in of 1,260 trading days (five years at 252 a year).
6. *10% drawdown in next quarter*: a fall of at least 10% within the next 63 trading days.
7. *Negative next-quarter return*: the market lower 63 trading days later.
8. *Jump model, high-risk state*: the two-sided state sequence of a two-state statistical jump model with $\lambda = 40$, fitted on the data from 1927 on. Its features are exponentially weighted means and downside deviations of daily returns, with half-lives of 10 and 42 days.
9. *NBER recession*: months the NBER dates as recession, without any announcement lag. The study uses it only to compare with the others.

Classifiers. The classifiers are logistic regressions (§5.4) with a ridge penalty of $\alpha = 0.1$ per training row on standardised inputs. The study refits them every January from 2000 to 2025. Each refit uses all rows since January 1990 whose label was final by the refit date. For forward labels, these are the rows older than the label's reach. For the ex-post bear label, they are the rows up to the last confirmed extreme. The study compares four input sets. *Vol-only* is the log of trailing 21-day realised volatility, the stale label for the volatility target. *VIX-only* is the log VIX. *Price* has seven price-path features. Three are log realised volatility over 5, 21 and 63 days. The other four are the drawdown from the one-year high, the 12-month and 1-month log returns, and the log distance from the 200-day average. *Price + cross-asset*, the full model, adds five more. These are the log VIX, a variance-risk-premium proxy, the Baa spread, its 21-day change, and the 10-year minus 3-month spread. The proxy is the log VIX minus log realised volatility.

Scores. The study reports AUC and the Brier skill score (§5.1, §7.1, §7.3) on the 2000–2025 test predictions. Climatology is the base rate in each refit's training data. An AUC of 0.5 is chance. A Brier skill of 0 is no better than climatology. Differences between models carry 90% intervals from a bootstrap that resamples one-year blocks of the test period.

6.2 Eight definitions, one century: how much they agree

The study compares the labels on the days where all of them are defined, October 1930 to October 2025 (24,809 days). The 1,260-day burn-in of the volatility threshold sets the start. The 63-day forward windows set the end. The ±20% rule finds 31 bear markets since 1926. The first ends before the common sample begins, so the table counts 30. For the same reason, the table counts 45 of the 47 bear markets of the ±15% rule.

Label	Share of days in the bad class	Episodes	Median length (days)	Reach
Bear, ±20% rule, ex post	19%	30	128	confirmation lag, median 70 at bear onsets
Bear, ±20% rule, as confirmed	14%	30	93	0
Bear, ±15% rule, ex post	20%	45	71	confirmation lag, median 48 at bear onsets
10% below the high, now	40%	132	3	0
Next-month volatility, top fifth	17%	166	16	21 days
10% drawdown in next quarter	21%	71	60	63 days
Negative next-quarter return	32%	501	4	63 days
Jump model, high-risk state	24%	52	68	whole sample; about eight days in practice
NBER recession	16%	15	209	months to years

The next table gives Cohen's kappa for each pair of labels, on the same days. Kappa corrects the raw share of agreement for the agreement that two labels with these base rates would reach by chance (Cohen, 1960):

$$\kappa = \frac{\text{agreement observed} - \text{agreement by chance}}{1 - \text{agreement by chance}}.$$

Two labels with base rates π_A and π_B agree by chance on a share $\pi_A\pi_B + (1-\pi_A)(1-\pi_B)$ of days. A kappa of 1 is perfect agreement. A kappa of 0 is no better than chance.

	1	2	3	4	5	6	7	8	9
1. Bear, ±20%, ex post	1.00								
2. Bear, ±20%, as confirmed	0.27	1.00							
3. Bear, ±15%, ex post	0.88	0.18	1.00						
4. 10% below the high	0.33	0.38	0.32	1.00					
5. Next-month volatility	0.35	0.24	0.33	0.21	1.00				
6. Drawdown next quarter	0.58	0.21	0.54	0.23	0.38	1.00			
7. Negative next quarter	0.38	0.03	0.38	0.08	0.11	0.52	1.00		
8. Jump model	0.50	0.47	0.45	0.47	0.50	0.37	0.11	1.00	
9. NBER recession	0.30	0.30	0.29	0.25	0.27	0.24	0.08	0.33	1.00

Four readings:

1. **Definitions in different families agree only moderately.** Apart from the ±20% and ±15% versions of the ex-post bear label, no pair reaches 0.6. Reasonable definitions do not converge on a single regime history.
2. **Thresholds matter less than hindsight.** The ±20% and ±15% ex-post labels agree at 0.88. The ±20% ex-post label and its own real-time version agree at 0.27.
3. **The ex-post bear label is mostly about the future.** It agrees better with *future* outcomes than with any rule that uses only the past. The kappa is 0.58 with a 10% drawdown in the next quarter, 0.38 with a negative next quarter and 0.35 with high volatility next month. It is 0.27 with its own real-time version and 0.33 with 10% below the high. The real-time version agrees with a negative next quarter at 0.03, which means no agreement.
4. **The jump model agrees with most labels at a middling level.** It agrees at about 0.5 with almost every label. It agrees with the direction label at 0.11, as a volatility state should.

Figure 1 shows eight of them, all but the ±15% variant, from 1995 to 2025.

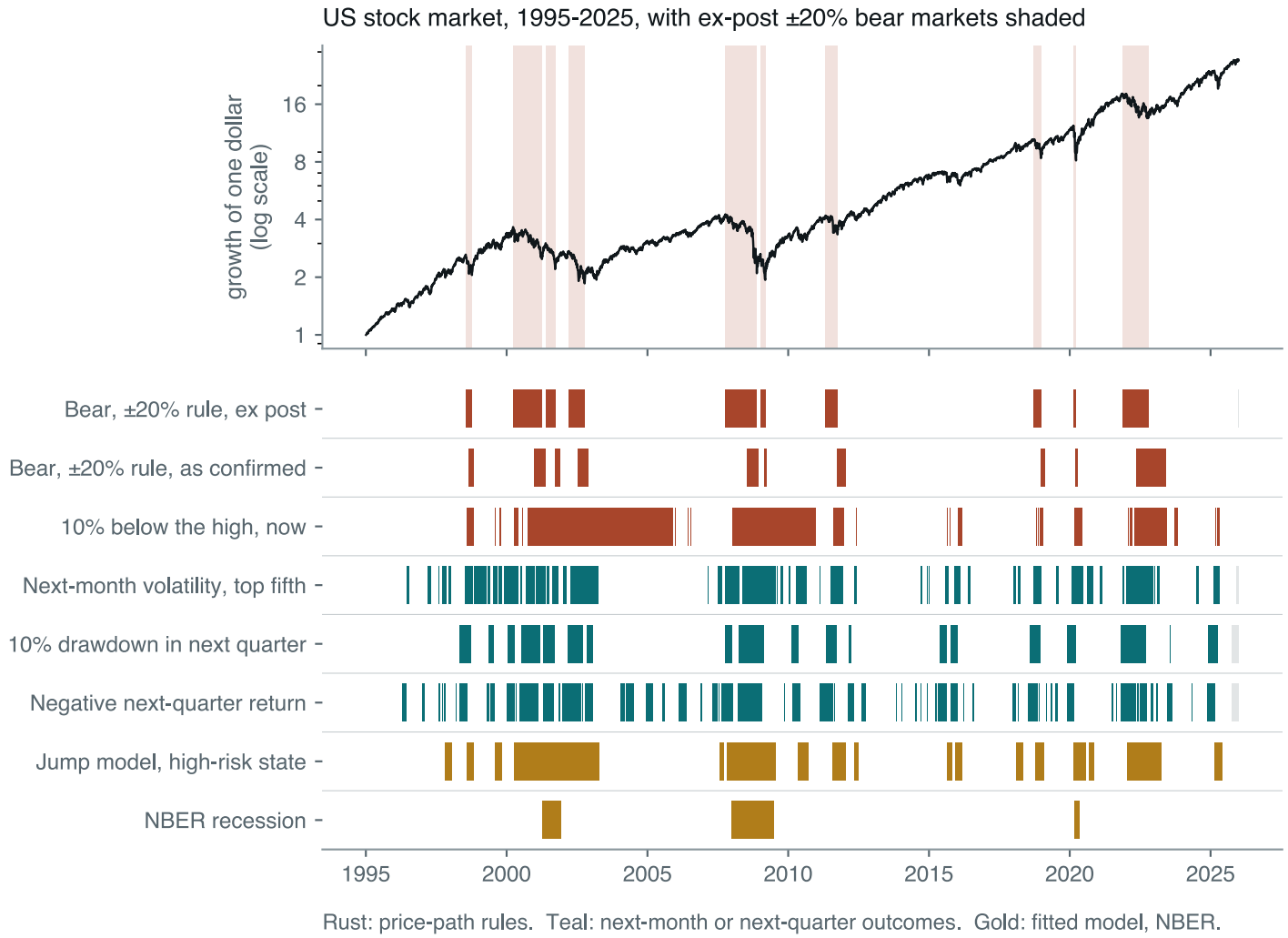


Figure 1. Eight definitions of a bad regime, 1995–2025. Rust strips show rules on the price path. Teal strips show outcomes over the next month or quarter. Gold strips show a fitted model and the NBER. Grey at the right edge marks the final weeks, where forward labels are not yet defined.

6.3 The price of hindsight: the ±20% rule

The ±20% rule is the most widely quoted regime definition. The comparison between its real-time version and its ex-post version is exact.

Confirmation lags. Since 1926, the rule confirmed a bear market a median of 70 trading days after its peak (interquartile range 42–146). It confirmed a bull market a median of 41 trading days after its trough (16–76). With 15% thresholds, the medians fall to 48 and 25. On 20% of all days since 1927, the confirmed state and the ex-post label disagree. On 12.7% of days the label is bear while the confirmed state still says bull. On 7.4% the reverse holds.

The bear markets since 2000 show the range of confirmation lags:

Peak	Confirmed	Trading days to confirmation	Fall at confirmation
24 Mar 2000	20 Dec 2000	188	21.9%
21 May 2001	17 Sep 2001	78	21.1%
19 Mar 2002	10 Jul 2002	78	20.2%
9 Oct 2007	15 Jul 2008	192	20.3%
6 Jan 2009	27 Feb 2009	36	20.4%
29 Apr 2011	3 Oct 2011	108	20.3%
20 Sep 2018	24 Dec 2018	65	20.4%
19 Feb 2020	12 Mar 2020	16	27.5%
8 Nov 2021	19 May 2022	133	20.0%

In 2011 and 2018, the rule confirmed the bear market on the day of its trough. So it recognised the bear market on the day it ended. The 2020 bear market fell 27.5% in the sixteen days the rule needed.

The label's value, and its timing. The table compares three ways to hold the US market or cash (the risk-free rate) from 1927 to 2025. The Sharpe ratio uses daily returns in excess of the risk-free rate. The maximum drawdown is the largest peak-to-trough fall in wealth:

Strategy	Compound annual return	Sharpe ratio	Maximum drawdown
Buy and hold	9.7%	0.45	84.1%
Out of the market in ex-post bear markets (the oracle)	24.2%	1.54	19.6%
Out of the market when the rule has confirmed a bear market	8.7%	0.46	61.3%

The oracle needs to know on each day whether that day belongs to a bear market. That is known only after confirmation. A real system must use the rule's state from the previous day. That gives a Sharpe ratio of 0.46, against 0.45 for buy and hold. The real-time rule does reduce the worst drawdown, from 84% to 61%, because it exits the deepest bear markets after their first 20%. The difference between 1.54 and 0.46 is the value of the unknowable 20% of days. This is the entire value available to a bull–bear classifier. A classifier can capture it only by forecasting falls of 20% before they have happened.

6.4 What a causal classifier learns

The study trains the classifiers on four of the labels. These are next-month volatility, a 10% drawdown in the next quarter, a negative next-quarter return and the ex-post ±20% bear label. Their future parts differ. The first label's future part is volatility. The second's is volatility and direction. The third's is direction. The fourth's is the current drawdown plus the direction of the coming months. The table gives the out-of-sample scores for 2000–2025 as AUC, with the Brier skill score against climatology in brackets:

Target	Base rate	Vol-only	VIX-only	Price	Price + cross-asset
Next-month volatility, top fifth	0.32	0.84 (+0.34)	0.87 (+0.38)	0.87 (+0.45)	0.88 (+0.47)
10% drawdown in next quarter	0.24	0.62 (+0.09)	0.61 (+0.08)	0.67 (+0.10)	0.64 (+0.07)
Negative next-quarter return	0.32	0.43 (−0.00)	0.39 (−0.00)	0.45 (−0.04)	0.44 (−0.08)
Bear, ±20% rule, ex post	0.19	0.71 (+0.15)	0.69 (+0.13)	0.83 (+0.27)	0.81 (+0.27)

The base rate is the share of test days in the bad class. For the volatility label it is 0.32, above the 17% of §6.2. The threshold is an expanding quantile, and the years since 2000 were more volatile than the history before them.

The next table gives the 90% bootstrap intervals for the gain of the full model over the price-only model:

Target	Gain in AUC	Gain in Brier skill
Next-month volatility, top fifth	[+0.001, +0.025]	[−0.003, +0.039]
10% drawdown in next quarter	[−0.085, −0.006]	[−0.077, −0.003]
Negative next-quarter return	[−0.042, +0.028]	[−0.089, −0.007]
Bear, ±20% rule, ex post	[−0.036, +0.002]	[−0.030, +0.034]

Against the volatility-only model, the full model's gain on the volatility target is [+0.023, +0.056] in AUC and [+0.10, +0.17] in Brier skill.

Five findings:

1. **Volatility labels are learnable, and mostly from persistence.** Trailing realised volatility alone forecasts next month's volatility regime with an AUC of 0.84. The VIX does better (0.87), as a forward-looking input should, and the full model adds about another 0.01.
2. **Direction labels are not learnable.** Every input set scores below 0.5 on the negative-next-quarter label, and none beats climatology on the Brier score. An AUC below 0.5 means that a loss followed the model's riskier days *less* often than its safer days. The link that the models learned between volatility and the next quarter's direction did not hold in the next period. The recoveries of 2009 and 2020 began near the peak of measured risk.
3. **Drawdown labels are partly learnable, through volatility.** A 10% drawdown is more likely when volatility is high, simply because a large fall is easier when the market moves more. The skill is real but small (Brier skill about 0.1), and the cross-asset inputs make it worse.
4. **The ex-post bear label is nowcastable, but only as far as the drawdown has already happened.** Price features reach an AUC of 0.83. The model largely detects the drawdown and the volatility that accompany a bear market already under way (§6.5).
5. **The cross-asset inputs add almost nothing on these targets.** The VIX, the variance risk premium, the credit spread and the yield curve improve the volatility forecast over trailing volatility alone. Over the seven price features, they add at most 0.01 of AUC on the volatility target. The 90% interval is +0.001 to +0.025, and the Brier-skill interval straddles zero. On the other targets, they make no difference or hurt. Two further checks point the same way. The study added the VIX/VIX3M term-structure ratio to models trained from September 2009 and tested them on 2015–2025. The volatility AUC stayed at 0.77, and the bear-label AUC fell from 0.78 to 0.76. A model *without* the term-structure ratio but trained from 1990 scored 0.80 and 0.81 on the same test years. Nineteen more years of history improved the scores more than the extra input did. With weak shrinkage ($\alpha = 0.001$), the twelve-input model's AUC on the drawdown label fell to 0.51, against 0.66 for the volatility-only model fitted the same way.

6.5 Detection at bear-market onsets

For the ex-post bear label, a practitioner asks how soon after each peak the classifier says "bear". The table counts trading days from each peak since 2000 to two events. The first is the rule's confirmation. The second is the first day the walk-forward classifier's probability reached one half.

Peak	Peak to trough (days)	Rule confirms	Price model	Price + cross-asset
24 Mar 2000	259	188	15	15
21 May 2001	82	78	4	6
19 Mar 2002	142	78	36	68
9 Oct 2007	283	192	69	31
6 Jan 2009	42	36	1	1
29 Apr 2011	108	108	67	68
20 Sep 2018	65	65	62	55
19 Feb 2020	23	16	12	6
8 Nov 2021	235	133	117	85

The classifiers cross one half well before the rule in the long bear markets of 2000 and 2007, by 120 to 170 days. They cross by 10 to 50 days before the rule in 2002, 2011 and 2021–22. They cross only days before the rule in the fast bear markets of 2018 and 2020. The first crossing overstates the classifier's lead, though. The crossings of one to six days in 2001 and 2009 are artefacts. The probability had not fallen far below one half during the rally before each peak. Over the whole test period, the full model's probability was above one half on 39% of ex-post bear days and on 6% of bull days. It gave 12 false alarms of five days or more. The confirmed state was on for 29% of bear days and 8% of bull days. The classifier is an earlier, noisier version of the rule.

Figure 2 shows why. The classifier's bear probability rises and falls with its forecast of volatility. It does not detect a bear-market rally such as November 2008 to January 2009, which the ±20% rule calls a bull market. It also stays high after the troughs: for weeks after March 2009, and less so after October 2022, when volatility remained elevated but the new bull market had begun.

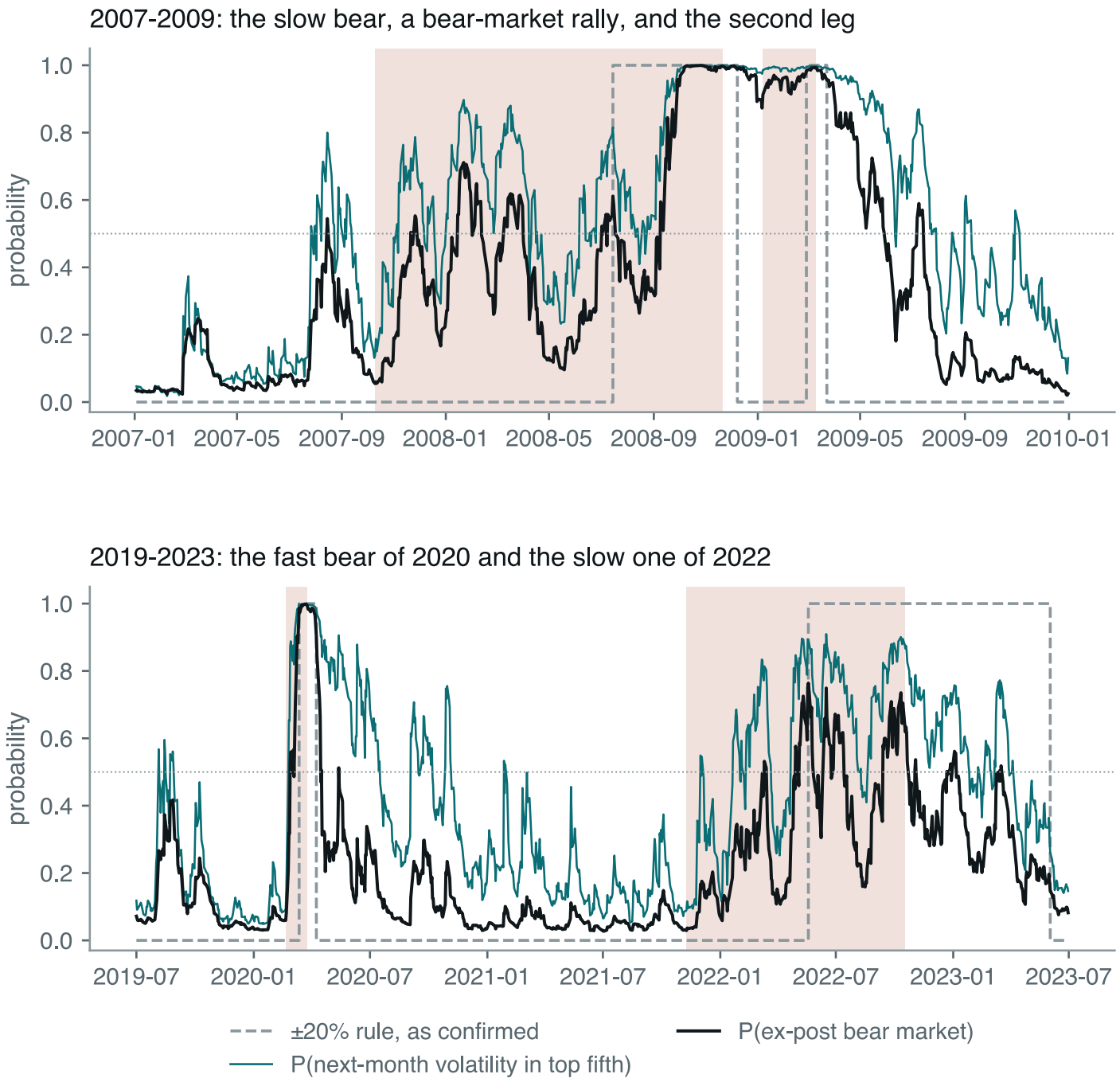


Figure 2. Walk-forward probabilities through the bear markets of 2007–09, 2020 and 2022. Shaded: ex-post ±20% bear markets. Black: the full model's probability of the ex-post bear label. Teal: its probability that next month's volatility falls in the top fifth. Dashed grey: the rule's confirmed state.

6.6 What the example does and does not show

The example supports the chapter's main argument in four ways. Labels whose future part is volatility were forecast well, mostly from persistence. Labels whose future part is direction were not forecast at all. The bull–bear label's value sat entirely in the days when it could not be known. The many inputs collapsed to the information already in the price path.

The example does not show that cross-asset inputs are useless. It has four limits:

1. **One market and one kind of label.** All the targets are about US equities. Labels about credit, the stock–bond correlation or a multi-asset portfolio would give credit spreads and the curve a direct role. The episodes of 2007 suggest that credit can lead (§2.11).

2. **A linear model.** The model included no interactions, such as high volatility together with widening spreads. A tree ensemble could find them, but then the few-episode problem applies.
3. **A short cross-asset history.** The cross-asset inputs enter in 1990, so the classifier's first refit, in January 2000, had seen two bear markets. The term-structure ratio starts in September 2009, so the first refit that used it, in January 2015, had seen one.
4. **Nine bear onsets.** Every statement about detection in §6.5 rests on nine episodes. The bootstrap intervals in §6.4 resample years, not episodes. They therefore understate the uncertainty about bear-market detection.

Within those limits, start a regime study with volatility labels and price-path inputs. Add the VIX. Add any other input only when a decision-specific label shows that it beats that baseline out of sample.

§6 Key takeaways

1. Eight reasonable definitions of a bad regime agree only moderately. Apart from the ±20% and ±15% versions of the ex-post bear label, no pair reaches a kappa of 0.6.
2. The ±20% bull–bear label is worth a Sharpe ratio of 1.54 known in advance and 0.46 known when confirmed, against 0.45 for buy and hold. Its value lies in the 20% of days it cannot be known.
3. The models forecast next month's volatility regime with an AUC near 0.88. Trailing volatility alone reaches 0.84.
4. No input set forecasts next quarter's direction. Every input set scored below 0.5 out of sample.
5. A bear-market classifier reads the drawdown and the volatility already present. It is an earlier, noisier version of the confirmation rule.
6. The VIX, credit spreads, the yield curve and the volatility term structure add at most 0.01 of AUC to price-path features for these equity labels. Longer history was worth more than an extra input.

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7. Evaluating a regime classifier

A regime classifier produces a daily probability, and a rule turns that probability into actions. The evaluation therefore has three layers. The first layer is the probability: whether it is accurate and calibrated. The second is the events: whether the classifier catches regime changes in time, and how often it raises false alarms. The third is the decision: whether acting on the classifier pays after costs. Each layer needs its own metric and the same discipline about uncertainty. The sample is a few dozen episodes, however many days it spans. The companion chapter's research protocol applies in full (section 11 of Market Regimes and Hidden Markov Models). This section adds what is specific to classifiers.

7.1 Proper scores and skill

A **proper scoring rule** has its best expected value when the forecast probability equals the true probability. Shading forecasts toward 0 or 1 therefore cannot improve the score (Gneiting & Raftery, 2007). Two proper rules are standard. The **Brier score** is the mean squared error of the probability over the n scored days,

$$\mathrm{BS} = \frac{1}{n}\sum_{t=1}^{n}\left(\hat{p}_t - y_t\right)^2,$$

(Brier, 1950). The **log loss**, $-\frac{1}{n}\sum_t \left[y_t \log \hat{p}_t + (1-y_t)\log(1-\hat{p}_t)\right]$, punishes confident errors far more heavily. The Brier score is the better default for regime work. A single confident miss in a crisis should count but should not dominate.

Raw scores depend on the base rate, so report **skill** against a reference forecast:

$$\mathrm{BSS} = 1 - \frac{\mathrm{BS}_{\text{model}}}{\mathrm{BS}_{\text{reference}}}.$$

A score of 0 means no better than the reference, 1 means perfect, and negative values mean worse. Use two references (§5.1). **Climatology** is for comparison with other studies. The **stale label** is for the decision. In §6.4, with the best input set for each label, the next-month volatility label reached a Brier skill of 0.47 over climatology. The drawdown label reached 0.10, and the direction label had a negative skill.

7.2 Calibration

A probability is **calibrated** if the label is positive on about 30% of the days on which the forecast is 30%. Check it with a reliability table. Group the forecasts into bins and compare each bin's average forecast with its observed frequency. The Brier score splits into three parts, reliability minus resolution plus uncertainty (Murphy, 1973). Index the bins by k. Bin k holds n_k days, with mean forecast $\bar{f}_k$ and observed frequency $\bar{y}_k$. Let $\bar{y}$ be the base rate over all n days. Then

$$\text{BS} = \underbrace{\sum_k \tfrac{n_k}{n}\,(\bar{f}_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\sum_k \tfrac{n_k}{n}\,(\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}\,(1-\bar{y})}_{\text{uncertainty}}.$$

The identity is exact when all the forecasts in a bin are the same number. For continuous forecasts, a small within-bin remainder remains. *Reliability* measures miscalibration and should be small. *Resolution* measures how far the bins' observed frequencies depart from the base rate. It shows how well the forecasts separate good days from bad, and it should be large. *Uncertainty* is the base rate's own variance. No model can change it.

Fit the calibration on training data only. For a flexible model, use out-of-fold scores (§5.11). Refit the calibration together with the model. Check calibration separately in calm and stressed periods. A model can be calibrated on average and badly miscalibrated in exactly the periods that matter.

7.3 Discrimination when the class is rare

The **AUC** is the area under the receiver operating characteristic curve. The curve plots the share of positive days flagged against the share of negative days flagged as the threshold moves. The AUC equals the probability that a randomly chosen positive day receives a higher score than a randomly chosen negative day. It ignores calibration entirely, because any monotone transformation of the scores leaves it unchanged. It measures ranking and nothing else. To get a rough scale, suppose the scores of positive and negative days were normal with equal spread and means Δ standard deviations apart. A positive score minus a negative score is then normal with mean Δ and variance 2. So AUC $= \Phi(\Delta/\sqrt{2})$, with Φ the standard normal distribution function. An AUC of 0.84 means Δ is about 1.4. An AUC of 0.6 means about 0.36.

When the class of interest is rare, as crisis regimes are, the AUC can look high while most of the alarms are false. In that case the **precision–recall curve** is more informative (Saito & Rehmsmeier, 2015). It plots the share of alarms that are correct against the share of positives caught. Report precision at the threshold the decision will actually use.

7.4 Event metrics: delay, false alarms and flicker

Scores averaged over days ignore the main job of a regime classifier, which is catching changes. Compute four event metrics per episode:

1. **Detection delay.** Days from the start of each positive episode to the first day the probability crosses the decision threshold. Report the median and the worst case. Compare them with the stale label's delay. For the ex-post bear label in §6.5, the price-feature classifier's delay ranged from 1 to 117 trading days. The shortest delays are crossings that never reset after the previous bear market, as in 2009. The confirmation rule's delay ranged from 16 to 192 trading days.
2. **False alarms.** Runs of at least m days above the threshold while the label is negative, per year. Set m by the decision. A sizing rule may tolerate one-day blips that a strategy switch cannot.
3. **Missed episodes.** Positive episodes never detected before they end.
4. **Flips per year.** The number of regime changes the post-processed output makes. Each flip has a trading cost. The companion chapter's cube-root rule for hysteresis bands applies (section 12 of Market Regimes and Hidden Markov Models).

Delay and false alarms trade against each other as the threshold moves. Plot the whole trade-off rather than one point on it. This trade-off is the event-level analogue of the receiver operating characteristic. The literature on monitoring for changes in behaviour treats it the same way (Fawcett & Provost, 1999).

7.5 Economic evaluation

The final test is the decision itself. Compare what acting on the classifier earns, after costs, against two alternatives. The first is the stale-label rule, which needs no model. The second is the continuous forecast that the regime coarsens (§5.10), such as sizing inversely to forecast volatility without any regime. A regime classifier that beats neither is not worth its complexity. In the worked example, the real-time ±20% rule, used for timing, earned what buy and hold earned (§6.3). Measure the classifier's own timing value the same way, including the cost of its false alarms. Map the probability to a position by a stated rule. Apply it with a one-day lag. Charge a cost for every change of position. Compare the Sharpe ratio and maximum drawdown with those of the two alternatives.

7.6 Uncertainty: count episodes, not days

Three tools measure uncertainty. They are listed in increasing order of strength:

1. **Block bootstrap.** Resample blocks of the test period. Make each block long enough to contain whole episodes, a year or more for bear markets. §6.4 gives an example. The stationary bootstrap is a variant with random block lengths (Politis & Romano, 1994). Intervals from day-level resampling are far too narrow.
2. **A persistence-preserving permutation test.** Break the link between the features and the label while keeping the persistence of each. Circularly shift the label sequence against the features by random offsets of at least a year. Refit and score. The distribution of scores under this null shows how much skill persistence and overlap alone produce.
3. **A synthetic null.** Simulate price paths from a model with no regimes, such as a GARCH process fitted to the same data. Compute the same labels and run the whole pipeline. Any "skill" it finds in direction labels is an artefact of the pipeline.

Count the trials. Every label definition, threshold, horizon and input set tried is a trial. The best of many trials overstates skill (Bailey & López de Prado, 2014; Harvey, Liu & Zhu, 2016).

7.7 A protocol

1. Write the decision, label, horizon and reach in one sentence (§1.2).
2. Vet the label: decision relevance, episodes, stability, reach, and the oracle and stale-label values (§4.9).
3. Build point-in-time inputs and a walk-forward training set that uses only labels final at each refit date (§4.10).
4. Fit the stale-label baseline and a regularised logistic model before anything else (§5.1, §5.4).
5. Score probabilities with the Brier score against climatology and against the stale label, check calibration, and report AUC with precision at the decision threshold (§7.1–§7.3).
6. Score events: delay, false alarms, misses and flips, per episode (§7.4).
7. Score the decision after costs against the stale-label rule and the continuous alternative (§7.5).
8. Put intervals on every comparison with a block bootstrap, run the permutation and synthetic nulls, and count the trials (§7.6).

§7 Key takeaways

1. Score probabilities with a proper rule, the Brier score by default, as skill over climatology and over the stale label.
2. Check calibration in calm and stressed periods separately.
3. AUC measures ranking only. For rare crisis classes, report precision at the threshold you will use.
4. Score events, not just days: detection delay, false alarms, misses and flips, per episode.
5. The decision test compares the classifier with the stale-label rule and with the continuous forecast it coarsens, after costs.
6. Resample whole episodes, test against nulls that keep persistence, and count every label, threshold and input set tried.

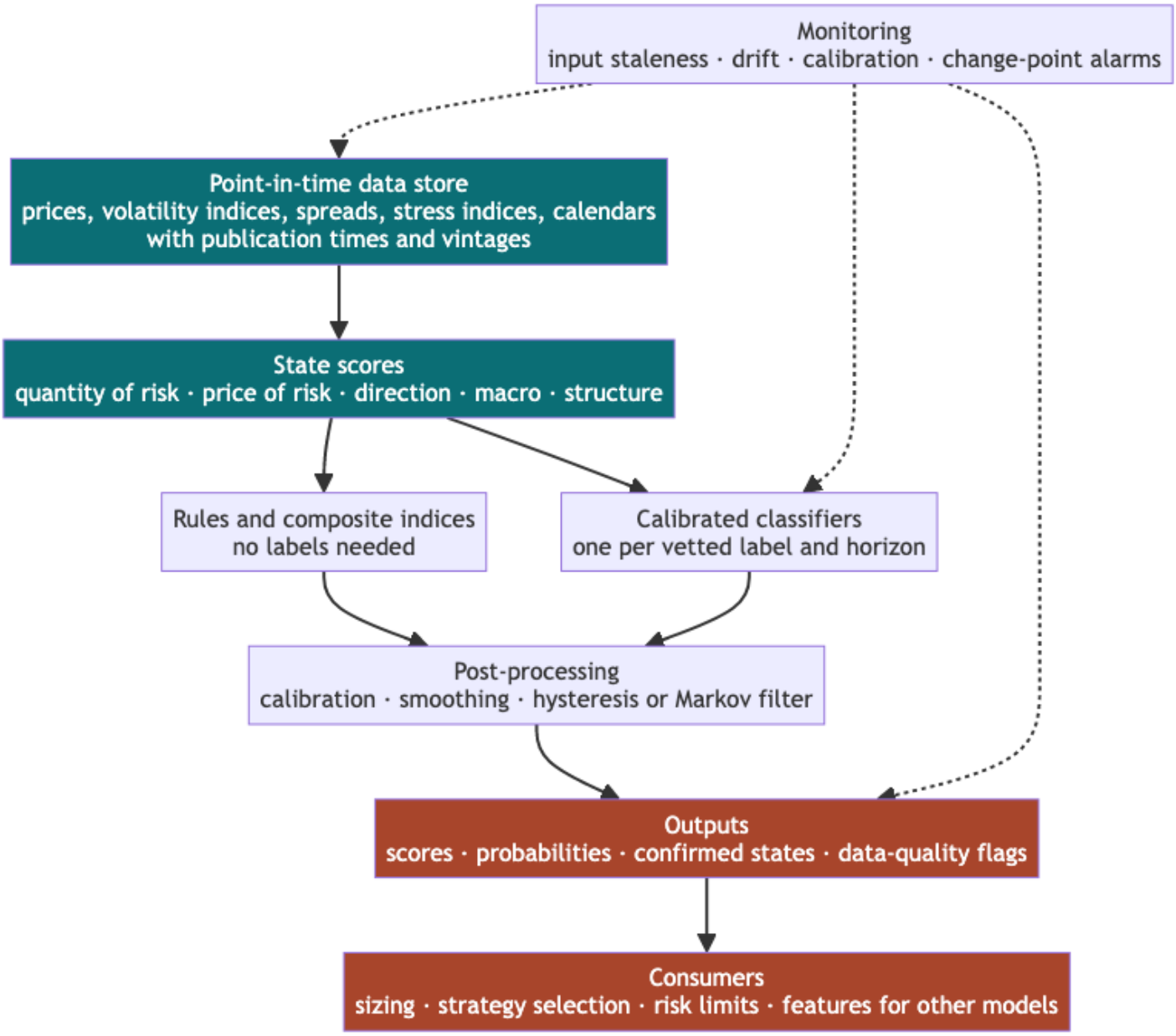
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8. Building a regime study

This section turns the chapter's results into an architecture. The architecture is generic. It assumes a daily data store, a study framework that computes derived series from inputs, and consumers such as position sizing. The main recommendation is that the study produces continuous scores and calibrated probabilities and derives discrete regimes from them, rather than producing a single regime label.

8.1 A reference architecture

The study has five layers. Data flow through them in one direction. A monitoring layer checks the data, the classifiers and the outputs.



The data store holds every input with the time it became available and, for revised series, every vintage (§3.11). Studies read only what was available at each date.

The state scores are a few continuous, causally standardised composites, with one or two per state variable (§3.13). One example is a quantity-of-risk score built from realised volatility at several horizons and the VIX. Another is a price-of-risk score built from credit spreads and the variance risk premium. The scores are also useful outputs on their own.

The regime models are of two kinds. Rules and composite indices need no labels and produce regimes with zero reach. The study trains one classifier per vetted label and horizon. Each classifier produces probabilities.

Post-processing calibrates the probabilities and smooths them into regimes (§5.11).

The outputs go to consumers. Each consumer uses the output that suits it. Sizing uses the risk score or the volatility probability. A strategy switch uses a smoothed regime. A risk limit uses a rule.

8.2 What the study should output

For each date, a regime study should publish:

1. **The state scores**, as continuous values.
2. **A calibrated probability for each label**, tagged with the label's definition and horizon. "P = 0.7" is meaningless without "that next month's realised volatility exceeds its historical 80th percentile".
3. **The stale label or confirmed state**, so that consumers and monitors can see the baseline the model is supposed to beat.
4. **The discrete regime**, derived from the probability by a stated rule, never as the only output.
5. **Data-quality flags**: which inputs were stale, missing or filled.
6. **The model version and training window**, so that anyone can reproduce any output.

A study that outputs only a regime label discards the probability, the uncertainty and the reason. Its consumers then cannot size by confidence.

8.3 A staged build with gates

The early stages are infrastructure. Skipping them is the usual reason regime studies fail in production. Each stage has a gate. The study must pass the gate before the next stage begins.

Stage	Build	Gate
0. Data	Point-in-time store; publication lags; vintages for revised series	An audit shows no input is used before its publication time
1. Scores and rules	State scores; rule-based regimes with hysteresis; stale labels	Scores and rules behave as expected in known episodes (2008, 2020, 2022)
2. Labels	Candidate labels for the decision; the five vetting checks	At least one label passes all five checks (§4.9)
3. Classifier	Regularised logistic model on five to ten inputs, walk-forward	Brier skill over the stale label, with a block-bootstrap interval above zero
4. Decision	The decision rule using the calibrated probability	Beats the stale-label rule and the continuous alternative after costs
5. Production	Monitoring, retraining schedule, fallbacks	Drift and staleness alarms tested by injecting stale data

Most studies should stop at stage 1 or 2. A set of well-built state scores and rules, with hysteresis, gives most of what a regime study can give for risk decisions. Stage 3 is worth building only where a vetted label shows a gap that a model can close.

8.4 Failure modes in production

- **Stale or missing inputs.** A source stops updating, or a holiday calendar differs between sources. The study then silently forward-fills an old value. Flag inputs older than their expected publication lag. Degrade to rules that do not use them.
- **Revisions and backfills.** A vendor revises history. The next retraining then learns from data that were never available live. Store each value as fetched, with its fetch time.
- **Structural breaks in an input.** The composition of option volume changed the meaning of put/call ratios (§3.9). An index methodology change shifts a level. Monitor each input's distribution against its own history.
- **Label drift.** The base rate of a label changes, for example because an expanding threshold adapts only slowly. The volatility label of §6.2 marks 17% of days over the century but 32% of the days since 2000 (§6.4). Report the base rate in each training window.
- **Calibration decay.** Probabilities calibrated in training drift over time. Recheck the reliability table at every retraining.
- **Regimes that change meaning.** A retrained latent-state model or jump model swaps or redefines its states. Identify states by a fixed property and alarm on large label revisions.
- **Crowding. [Hypothesis]** Many funds cutting risk on the same VIX threshold may amplify the moves they respond to. Prefer continuous sizing to hard thresholds. Continuous sizing also reduces the study's own trading.

§8 Key takeaways

1. Build in layers: point-in-time data, continuous state scores, rules and classifiers, post-processing, outputs, with monitoring across all of them.
2. Output scores and calibrated probabilities with their label definitions, the stale baseline, data-quality flags and versions. Never only a regime label.
3. Pass the gates in order. Data integrity and well-behaved state scores come before any classifier.
4. Most studies should stop at state scores and rules with hysteresis. Train a classifier only where a vetted label shows a gap a model can close.
5. In production, the failures are stale inputs, revisions, structural breaks in inputs and decaying calibration. Monitor all four.

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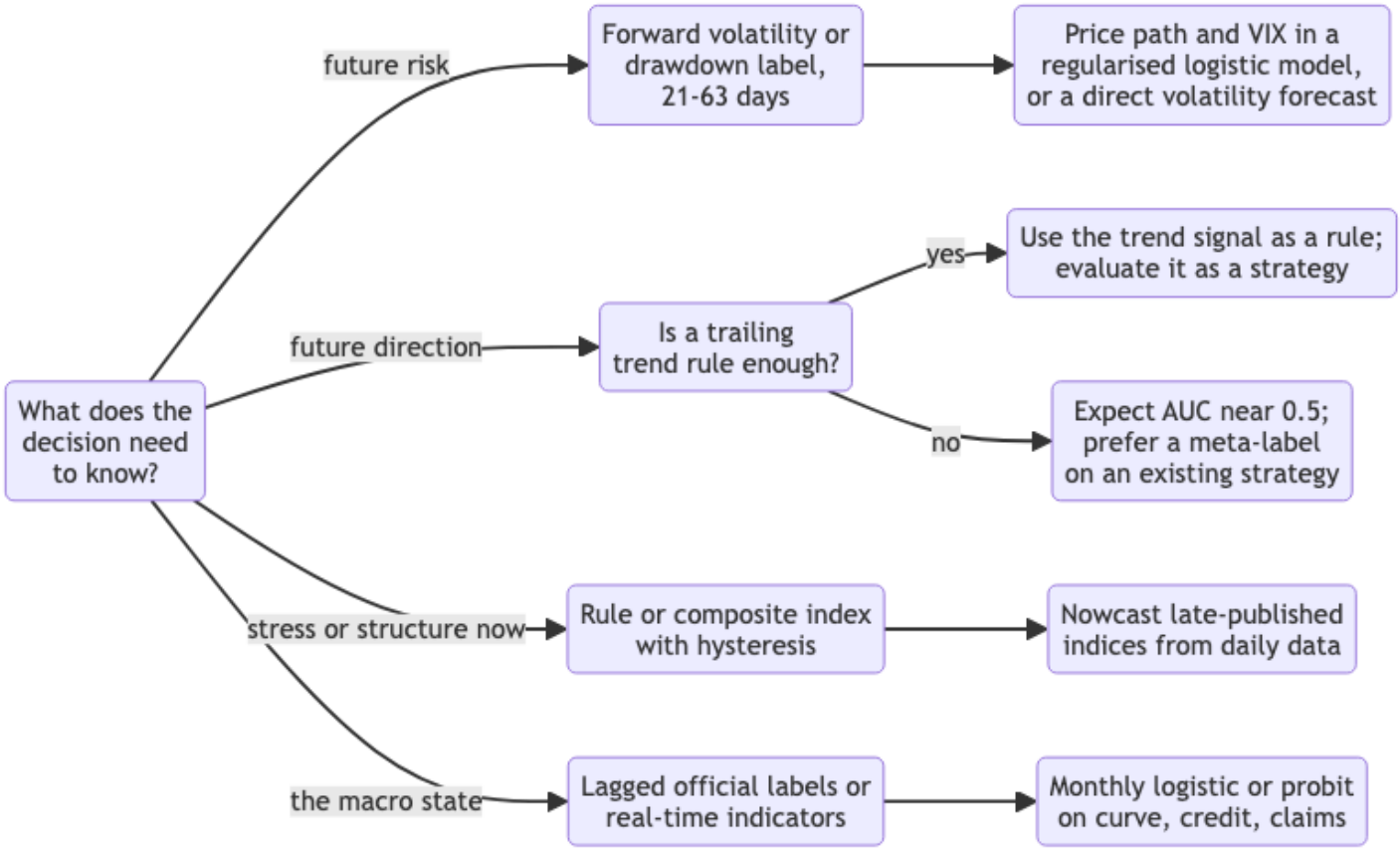
9. Synthesis

9.1 The framework in one paragraph

A regime is a coarse partition of a few continuous state variables, chosen to serve a decision. The state variables are the quantity of risk, the price of risk, direction, the macroeconomic state and structure. Inputs are noisy measurements of those state variables. Most inputs measure the quantity of risk. A label is a rule on a window of the price path. Its reach, how far into the future the window extends, decides everything else: the purge in training, the floor on detection delay, and how much a classifier can learn. A supervised regime classifier forecasts the future part of its label. When that future part is volatility, which persists, the classifier learns a great deal, mostly from the stale label. When it is direction, which does not persist, the classifier learns almost nothing. The label's value then lies in the days when it cannot be known. The sample size is the number of episodes, so models must be small and evaluation must resample episodes.

Decision	State variable	Label	Reach	First inputs	First model	Baseline to beat
Sizing, risk limits	quantity of risk	forward volatility quantile	21 days	realised volatility, VIX	logistic, or a direct volatility forecast	trailing volatility
Drawdown control	quantity of risk, direction	forward drawdown event	63 days	realised volatility, drawdown, VIX	logistic	trailing volatility
Exposure timing	direction	ex-post bull–bear or forward return	months	price path	none worth fitting	confirmed state; buy and hold
Leverage, liquidity	price of risk, structure	stress index above a threshold	0, plus publication lag	spreads, stress indices	rule with hysteresis; nowcast of the index	the last published value
Strategic allocation	macro	lagged recession or growth–inflation phase	months	curve, credit, claims, surprises	monthly logistic or probit	the last published phase
Diversification	structure	correlation regime	0 (estimated)	rolling correlations, implied correlation	rule with hysteresis	rolling estimate
Strategy selection	depends	meta-label of the base strategy	its holding period	the strategy's own state plus risk inputs	logistic	always on

9.2 Decision tree



The same steps come first on every branch. Write the decision as a sentence. Vet the label (§4.9). Build the stale-label baseline. Align every input to its publication time. Count episodes rather than days.

9.3 Ten rules for someone starting today

1. Write the decision, the label, the horizon and the reach in one sentence before loading any data.
2. Prefer labels whose future part is volatility. They can be forecast. Direction cannot.
3. Compute the label's reach and purge accordingly. A label with zero reach is a rule, so compute it instead of learning it.
4. Beat the stale label, not the base rate.
5. Start with the price path and the VIX. Add any other input only when it beats that baseline out of sample on a decision-specific label.
6. Align every input to its publication time and store vintages of revised series.
7. Use expanding percentiles for levels. Keep rolling z-scores for surprises.
8. Use a regularised logistic model on five to ten inputs before anything more complex, and calibrate whatever you use.
9. Output continuous scores and calibrated probabilities. Derive regimes from them with hysteresis.
10. Count episodes, not days. Resample episodes, test against nulls that keep persistence, and count every trial.

9.4 What is known, and what is not

Established. **[Fact]** Volatility is persistent and forecastable from its own history and from implied volatility. So regimes defined by future risk can be forecast with real skill. **[Fact]** Official macroeconomic regime labels arrive months to years late and are not observable in real time.

Contested. Researchers disagree on three questions. The first is whether regimes defined by future direction can be forecast with useful skill from public market data. Most proposed predictors of the equity premium fail out of sample (Welch & Goyal, 2008), and the worked example finds the same for regime labels. Forecasts with sensible restrictions (Campbell & Thompson, 2008) or combinations of many predictors (Rapach, Strauss & Zhou, 2010) recover small but real gains. The disagreement is about whether small means useful. No one finds the large skill a direction regime would need. The second question is whether cross-asset inputs (credit, the curve, the volatility term structure, sentiment) add to price-path inputs for equity regimes. The worked example did not test sentiment. It finds a gain of at most 0.01 of AUC for its labels. Practitioners report

gains in credit-led episodes such as 2007, and teacher–student studies report gains for multi-asset allocation. Both sides rest on few episodes. The third question is whether volatility timing improves returns, rather than only risk, out of sample (Moreira & Muir, 2017; Cederburg et al., 2020).

Open. Three questions are open. The first is whether labels defined directly on multi-asset outcomes, such as the stock–bond correlation or a stress event, are forecastable from structure inputs with useful skill. The second is how to share episodes across markets. A classifier trained on many countries' bear markets would have more episodes to learn from than one trained on the US alone. The third is whether the composition changes in option markets since 2020 have created new information in sentiment inputs or destroyed the old.

This is the chapter's view, stated as such. A regime study is valuable as a disciplined measurement of the quantity and price of risk, with added probabilities for a few risk-type labels. The search for a classifier that tells bull markets from bear markets early is the search for a return forecast. Judge it as one.

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10. References

Entries fall into groups by kind. Each entry has one line on why it matters. Practitioner sources name the author's interest. Critiques have their own group. Links go to a free copy where one exists. Entries marked paywalled link to the publisher.

10.1 Regime definitions and dating

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- **OECD (2012).** *OECD System of Composite Leading Indicators.* Prepared by G. Gyomai and E. Guidetti. — How to build the composite leading indicators, including the Hodrick–Prescott detrending and Bry–Boschan dating.
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10.2 Inputs

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- **Favara, G., Gilchrist, S., Lewis, K. F. & Zakrajšek, E. (2016).** "Updating the Recession Risk and the Excess Bond

Premium." FEDS Notes. — The Federal Reserve's maintained version of the series.
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- **Estrella, A. & Mishkin, F. S. (1998).** "Predicting U.S. Recessions: Financial Variables as Leading Indicators." *Review of Economics and Statistics* 80(1), 45–61. — The term spread beats other financial variables beyond one quarter.
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- **Adrian, T., Boyarchenko, N. & Giannone, D. (2019).** "Vulnerable Growth." *American Economic Review* 109(4), 1263–1289. — Financial conditions forecast the lower tail of growth, not its median.
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- **Simon, D. P. & Wiggins, R. A. (2001).** "S&P Futures Returns and Contrary Sentiment Indicators." *Journal of Futures Markets* 21(5), 447–462. [publisher — paywalled] — The VIX, put/call and TRIN as contrarian signals.
- **Pan, J. & Poteshman, A. M. (2006).** "The Information in Option Volume for Future Stock Prices." *Review of Financial Studies* 19(3), 871–908. — Option volume predicts returns through non-public open-buy volume, not the public ratios.
- **Bryzgalova, S., Pavlova, A. & Sikorskaya, T. (2023).** "Retail Trading in Options and the Rise of the Big Three Wholesalers." *Journal of Finance* 78(6), 3465–3514. [publisher — paywalled] — The retail option boom that changed what option-volume ratios measure.
- **Cboe (2025).** "SPX 0DTE Options Jump to 61% Share on Retail Resurgence." Cboe Insights. — The share of same-day expiries in S&P 500 option volume. The exchange profits from this volume.
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- **Scotti, C. (2016).** "Surprise and Uncertainty Indexes: Real-Time Aggregation of Real-Activity Macro-Surprises." *Journal of Monetary Economics* 82, 1–19. [publisher — paywalled] — How to build a macroeconomic surprise index.
- **Croushore, D. & Stark, T. (2001).** "A Real-Time Data Set for Macroeconomists." *Journal of Econometrics* 105(1), 111–130. [publisher — paywalled] — Why vintages matter, and the data set that makes them available.
- **Giannone, D., Reichlin, L. & Small, D. (2008).** "Nowcasting: The Real-Time Informational Content of Macroeconomic Data." *Journal of Monetary Economics* 55(4), 665–676. [publisher — paywalled] — Updating an estimate of the present as each release arrives.

10.3 Labels, classifiers and machine learning

- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Triple-barrier labels, meta-labels, sample uniqueness, and purged and embargoed cross-validation.
- **López de Prado, M. (2020).** *Machine Learning for Asset Managers.* Cambridge Elements in Quantitative Finance. — Trend-scanning labels (chapter 5).
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- **Kim, C.-J. (1994).** "Dynamic Linear Models with Markov-Switching." *Journal of Econometrics* 60(1–2), 1–22. [publisher — paywalled] — The smoother whose output becomes a soft label. Used as a signal, it leaks future data.
- **Bemporad, A., Breschi, V., Piga, D. & Boyd, S. (2018).** "Fitting Jump Models." *Automatica* 96, 11–21. — The jump model and its fitting algorithm.
- **Nystrup, P., Lindström, E. & Madsen, H. (2020).** "Learning Hidden Markov Models with Persistent States by Penalizing

Jumps." *Expert Systems with Applications* 150, 113307. [publisher — paywalled] — Jump models for financial regimes.
- **Aydınhan, A. O., Kolm, P. N., Mulvey, J. M. & Shu, Y. (2024).** "Identifying Patterns in Financial Markets: Extending the Statistical Jump Model for Regime Identification." *Annals of Operations Research*. [DOI] — Continuous and probabilistic extensions of the jump model.
- **Horvath, B., Issa, Z. & Muguruza, A. (2024).** "Clustering Market Regimes Using the Wasserstein Distance." *Journal of Computational Finance*. — Clustering whole return distributions rather than feature vectors.
- **Botte, A. & Bao, D. (2021).** "A Machine Learning Approach to Regime Modeling." Two Sigma. — A Gaussian mixture on factor returns with four named conditions. The authors' firm sells factor analytics.
- **Shu, Y., Yu, C. & Mulvey, J. M. (2024a).** "Downside Risk Reduction Using Regime-Switching Signals: A Statistical Jump Model Approach." *Journal of Asset Management* 25(5), 493–507. — Jump-model regimes for drawdown control.
- **Shu, Y., Yu, C. & Mulvey, J. M. (2024b).** "Dynamic Asset Allocation with Asset-Specific Regime Forecasts." *Annals of Operations Research* 346(1), 285–318. — The teacher–student design: jump-model labels forecast by gradient-boosted trees.
- **Shu, Y. & Mulvey, J. M. (2024).** "Dynamic Factor Allocation Leveraging Regime-Switching Signals." *Journal of Portfolio Management* 51(3), 50–72. — The same design for factor allocation.
- **Hauptmann, J., Hoppenkamps, A., Min, A., Ramsauer, F. & Zagst, R. (2014).** "Forecasting Market Turbulence Using Regime-Switching Models." *Financial Markets and Portfolio Management* 28(2), 139–164. [publisher — paywalled] — An early teacher–student system: logistic regressions forecasting Markov-switching probabilities.
- **Uysal, A. S. & Mulvey, J. M. (2021).** "A Machine Learning Approach in Regime-Switching Risk Parity Portfolios." *Journal of Financial Data Science* 3(2), 87–108. [publisher — paywalled] — Random forests on macro data for recession and contraction probabilities.
- **Kauppi, H. & Saikkonen, P. (2008).** "Predicting U.S. Recessions with Dynamic Binary Response Models." *Review of Economics and Statistics* 90(4), 777–791. [publisher — paywalled] — Dynamic probit models for recessions.
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- **Chen, S.-S. (2009).** "Predicting the Bear Stock Market: Macroeconomic Variables as Leading Indicators." *Journal of Banking & Finance* 33(2), 211–223. [publisher — paywalled] — Which macro variables help predict bear markets.
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- **Filardo, A. J. (1994).** "Business-Cycle Phases and Their Transitional Dynamics." *Journal of Business & Economic Statistics* 12(3), 299–308. [publisher — paywalled] — The applied template for time-varying transitions.
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- **Chen, L., Pelger, M. & Zhu, J. (2024).** "Deep Learning in Asset Pricing." *Management Science* 70(2), 714–750. — Learned macroeconomic states inside an asset-pricing objective.
- **Hinton, G., Vinyals, O. & Dean, J. (2015).** "Distilling the Knowledge in a Neural Network." arXiv:1503.02531. — Training a student on a teacher's soft outputs.
- **Frénay, B. & Verleysen, M. (2014).** "Classification in the Presence of Label Noise: A Survey." *IEEE Transactions on Neural Networks and Learning Systems* 25(5), 845–869. [publisher — paywalled] — Which kinds of label noise classifiers tolerate.
- **Adams, R. P. & MacKay, D. J. C. (2007).** "Bayesian Online Changepoint Detection." arXiv:0710.3742. — The run-length posterior.
- **Breiman, L. (2001).** "Random Forests." *Machine Learning* 45(1), 5–32. [publisher — paywalled] — Averaged trees grown on bootstrap samples.
- **Friedman, J. H. (2001).** "Greedy Function Approximation: A Gradient Boosting Machine." *Annals of Statistics* 29(5), 1189–1232. — Gradient boosting.
- **Koenker, R. & Bassett, G. (1978).** "Regression Quantiles." *Econometrica* 46(1), 33–50. [publisher — paywalled] — Quantile regression, the basis of growth-at-risk.
- **Page, E. S. (1954).** "Continuous Inspection Schemes." *Biometrika* 41(1–2), 100–115. [publisher — paywalled] — The CUSUM change detector.
- **Bourlard, H. & Morgan, N. (1994).** *Connectionist Speech Recognition: A Hybrid Approach.* Kluwer. [publisher — paywalled] — Neural classifiers inside a hidden Markov model, the template for §5.11.
- **Niculescu-Mizil, A. & Caruana, R. (2005).** "Predicting Good Probabilities with Supervised Learning." *Proceedings of the 22nd International Conference on Machine Learning*, 625–632. — Which classifiers need calibration, and how.
- **Cohen, J. (1960).** "A Coefficient of Agreement for Nominal Scales." *Educational and Psychological Measurement* 20(1), 37–46. [publisher — paywalled] — Kappa.

10.4 Evaluation

- **Brier, G. W. (1950).** "Verification of Forecasts Expressed in Terms of Probability." *Monthly Weather Review* 78(1), 1–3. — The Brier score.
- **Murphy, A. H. (1973).** "A New Vector Partition of the Probability Score." *Journal of Applied Meteorology* 12(4), 595–600. — Reliability, resolution and uncertainty.
- **Gneiting, T. & Raftery, A. E. (2007).** "Strictly Proper Scoring Rules, Prediction, and Estimation." *Journal of the American Statistical Association* 102(477), 359–378. [publisher — paywalled] — The theory of proper scores.
- **Saito, T. & Rehmsmeier, M. (2015).** "The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets." *PLOS ONE* 10(3), e0118432. — Why rare crisis classes need precision and recall.
- **Fawcett, T. (2006).** "An Introduction to ROC Analysis." *Pattern Recognition Letters* 27(8), 861–874. [publisher — paywalled] — The standard tutorial on ROC curves and AUC.
- **Fawcett, T. & Provost, F. (1999).** "Activity Monitoring: Noticing Interesting Changes in Behavior." *Proceedings of the Fifth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 53–62. — Evaluating detectors of change by delay and false alarms.
- **Politis, D. N. & Romano, J. P. (1994).** "The Stationary Bootstrap." *Journal of the American Statistical Association* 89(428), 1303–1313. [publisher — paywalled] — Resampling dependent data in blocks.
- **Bailey, D. H. & López de Prado, M. (2014).** "The Deflated Sharpe Ratio." *Journal of Portfolio Management* 40(5), 94–107. — Adjusting for the number of trials.
- **Harvey, C. R., Liu, Y. & Zhu, H. (2016).** "…and the Cross-Section of Expected Returns." *Review of Financial Studies* 29(1), 5–68. [publisher — paywalled] — Multiple testing in finance.

10.5 Critiques and cautions

- **Welch, I. & Goyal, A. (2008).** "A Comprehensive Look at the Empirical Performance of Equity Premium Prediction." *Review of Financial Studies* 21(4), 1455–1508. — Most return predictors fail out of sample. This is the evidence behind "direction is not learnable".
- **Campbell, J. Y. & Thompson, S. B. (2008).** "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" *Review of Financial Studies* 21(4), 1509–1531. — The reply to Welch and Goyal. With simple restrictions, out-of-sample gains are small but positive.
- **Rapach, D. E., Strauss, J. K. & Zhou, G. (2010).** "Out-of-Sample Equity Premium Prediction: Combination Forecasts and Links to the Real Economy." *Review of Financial Studies* 23(2), 821–862. — Combining many weak predictors recovers a little out-of-sample skill.
- **Cederburg, S., O'Doherty, M. S., Wang, F. & Yan, X. S. (2020).** "On the Performance of Volatility-Managed Portfolios." *Journal of Financial Economics* 138(1), 95–117. [publisher — paywalled] — Out of sample, volatility management rarely beats the unmanaged strategy.
- **Moreira, A. & Muir, T. (2017).** "Volatility-Managed Portfolios." *Journal of Finance* 72(4), 1611–1644. — The case for volatility timing. Read it with Cederburg et al.
- **Brown, G. W. & Cliff, M. T. (2004).** "Investor Sentiment and the Near-Term Stock Market." *Journal of Empirical Finance* 11(1), 1–27. [publisher — paywalled] — Sentiment and breadth move with returns but predict little.
- **Wang, Y.-H., Keswani, A. & Taylor, S. J. (2006).** "The Relationships between Sentiment, Returns and Volatility." *International Journal of Forecasting* 22(1), 109–123. — Returns and volatility cause sentiment measures.
- **Hamilton, J. D. (2018).** "Why You Should Never Use the Hodrick-Prescott Filter." *Review of Economics and Statistics* 100(5), 831–843. — Two-sided filters and their end-point revisions.
- **Forbes, K. J. & Rigobon, R. (2002).** "No Contagion, Only Interdependence: Measuring Stock Market Comovements." *Journal of Finance* 57(5), 2223–2261. — Correlation rises mechanically with volatility.
- **Brown, S. J., Goetzmann, W. N., Ibbotson, R. G. & Ross, S. A. (1992).** "Survivorship Bias in Performance Studies." *Review of Financial Studies* 5(4), 553–580. [publisher — paywalled] — How conditioning on survivors distorts history.

10.6 Books

- **Williams, D. (1991).** *Probability with Martingales.* Cambridge University Press. — Conditional expectation and filtrations, rigorously.

- **Hamilton, J. D. (1994).** *Time Series Analysis.* Princeton University Press. — Autoregressions, persistence and regime-switching models.
- **Hastie, T., Tibshirani, R. & Friedman, J. (2009).** *The Elements of Statistical Learning*, 2nd ed. Springer. — Logistic regression, shrinkage, clustering, trees and boosting. Free from the authors.
- **Jolliffe, I. T. (2002).** *Principal Component Analysis*, 2nd ed. Springer. — The reference on principal components.

10.7 Practitioner sources

Practitioner frameworks are often the clearest statement of how regimes are used. They come from firms that sell products built on them.

- **Podolsky, P., Johnson, R. & Jennings, O. (2012).** "The All Weather Story." Bridgewater Associates. — Four growth–inflation environments and risk balance across them. The firm manages All Weather portfolios.
- **Greetham, T. and colleagues (2004).** *The Investment Clock.* Merrill Lynch. — Reflation, recovery, overheat and stagflation. No public copy located.
- **Faber, M. T. (2007).** "A Quantitative Approach to Tactical Asset Allocation." *Journal of Wealth Management* 9(4), 69–79. [publisher — paywalled] — The ten-month moving-average rule. The author's firm runs trend-following funds.

10.8 Data sources

- Kenneth R. French data library — daily US market and risk-free returns from 1926.
- Cboe VIX historical data — the VIX and its term-structure and related indices.
- FRED, Federal Reserve Bank of St. Louis — BAA10Y, T10Y3M, USREC, NFCI and STLFSI4. ALFRED provides vintages.
- Chicago Fed NFCI — the index, its subindices and documentation.
- OFR Financial Stress Index — daily, with category and regional contributions.

10.9 If you only read eight things

1. **Ang & Timmermann (2012)** — the review of regime models in finance.
2. **Kole & van Dijk (2017)** — rules against models for bull and bear markets.
3. **Lunde & Timmermann (2004)** — what a turning-point label is and how it behaves.
4. **López de Prado (2018)** — labels with reach, and validation that respects it.
5. **Bekaert & Hoerova (2014)** — what the VIX contains.
6. **Adrian, Boyarchenko & Giannone (2019)** — what financial conditions forecast, and what they do not.
7. **Welch & Goyal (2008)** — why direction labels forecast poorly.
8. **Gneiting & Raftery (2007)** — how to score a probability.

Appendix A. Concepts and prerequisites

This appendix covers the concepts the main text uses without explaining them. The entries follow **dependency** order, not alphabetical order, so later entries build on earlier ones. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to read more. Several entries summarise material that the companion chapter, Market Regimes and Hidden Markov Models, develops at length. These entries link to its appendix instead of repeating it.

Index. Where each concept is first used:

Concept	First used	Concept	First used
Information sets and look-ahead	§1.4	Clustering	§4.6
Conditional probability forecasts	§1.4	Trees and boosting	§5.5
Persistence and half-life	§1.4	Quantile regression	§5.10
Sampling error of correlations	§2.5	Hysteresis	§1.4
Proper scores and the Brier score	§1.4	Two-sided filters	§2.8

Concept	First used	Concept	First used
ROC curves and AUC	§1.4	Markov chains and durations	§5.11
Cohen's kappa	§1.1	Hidden Markov models and filtering	§1.5
Information coefficient and hit rate	§1.4	Statistical jump models	§4.6
Sharpe ratio and drawdown	§1.1	Change-point detection	§5.9
Logistic regression	§5.4	Knowledge distillation	§4.10
Shrinkage	§5.4	Realised volatility	§3.2
Calibration	§5.11	Implied volatility and the VIX	§2.3
Overlapping labels	§4.5	Credit spreads	§3.4
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Purging, embargo, walk-forward	§4.5	Stress and conditions indices	§2.7
Block bootstrap and permutation nulls	§6.1	Survivorship bias	§3.7
Multiple testing	§3.1	Variance ratio and Hurst exponent	§2.5
Principal components and factor models	§2.6		

A.1 Information sets and look-ahead

The idea. A backtest is valid only if every number it uses was available when the decision was made. The *information set* $\mathcal{F}_t$ records what was available: everything published by the close of day t, in the version published then. Look-ahead bias is the use of anything outside it.

Formally. $\{\mathcal{F}_t\}$ is a filtration: an increasing family of information sets, $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. A quantity is $\mathcal{F}_t$-measurable if its value is determined by $\mathcal{F}_t$. In practice, test whether anyone could have computed this number from data published by the close of day t, in the version that existed then. A revised series has a *vintage* for each publication date. Only the vintage current at t belongs to $\mathcal{F}_t$.

Why it appears here. The reach of a label (§1.4) determines which information set the label belongs to. The timing table of §3.11 lists when each input enters $\mathcal{F}_t$. Its rules keep features $\mathcal{F}_t$-measurable.

Deeper. Appendix A.1 of Market Regimes and Hidden Markov Models; Williams (1991), chapters 9 and 10; on vintages, Croushore & Stark (2001).

A.2 Conditional probability forecasts

The idea. For a binary label, the best possible forecast is the probability of 1 given everything known. It is not a guess of 0 or 1. Every classifier tries to estimate that probability.

Formally. $p_t = \Pr(y_t = 1 \mid \mathcal{F}_t) = \mathbb{E}[y_t \mid \mathcal{F}_t]$. Among all forecasts built from $\mathcal{F}_t$ it minimises the expected squared error. The minimum is $\mathbb{E}[\mathrm{Var}(y_t \mid \mathcal{F}_t)] = \mathbb{E}[p_t(1-p_t)]$, because a binary variable with success probability p has variance $p(1-p)$.

Why it appears here. It is the upper limit on any classifier's accuracy (§1.4). It also explains why a label with zero reach needs no model. For such a label, p_t is 0 or 1 and the upper limit is perfect accuracy.

Deeper. Williams (1991), chapter 9, on conditional expectation as the best mean-squared-error predictor.

A.3 Persistence and half-life

The idea. A persistent series stays near where it is. Persistence is why yesterday's regime predicts today's, why volatility can be forecast, and why daily samples contain far fewer independent observations than days.

Formally. Take a first-order autoregression $x_t = \mu + \rho\,(x_{t-1} - \mu) + e_t$ with $0 < \rho < 1$. A deviation from the mean shrinks by the factor ρ each period. After k periods it is ρ^k of its size. It halves after $\ln 0.5 / \ln \rho$ periods. With $\rho = 0.98$ per day, the half-life is about 34 days. A slope of 0.72 per month gives $\ln 0.5 / \ln 0.72 \approx 2.1$ months.

Why it appears here. The persistence of volatility makes volatility labels learnable and the stale label a strong baseline (§1.4, §5.1). The lack of persistence in returns makes direction labels flicker (§4.5).

Deeper. Hamilton (1994), chapter 3.

A.4 Sampling error of correlations

The idea. A correlation or autocorrelation estimated from a short window is noisy, and the noise can be as large as the variation the estimate is meant to detect.

Formally. For n independent pairs with true correlation ρ, the estimate's standard error is approximately $(1 - \rho^2)/\sqrt{n}$. At $\rho = 0$ this is $1/\sqrt{n}$, about 0.13 for $n = 63$. The approximation needs n moderately large and the pairs independent. Overlapping or persistent data make the error larger. Fisher's transformation $\frac{1}{2} \ln[(1 + r)/(1 - r)]$ gives an approximately normal statistic with standard error $1/\sqrt{n - 3}$. This statistic is the better basis for intervals.

Why it appears here. It limits trending-versus-mean-reverting regimes (§2.5) and correlation regimes (§2.9, §3.8).

Deeper. Any mathematical statistics text on the sampling distribution of the correlation coefficient.

A.5 Proper scores and the Brier score

The idea. A scoring rule grades a probability forecast against what happened. It is *proper* if no forecaster can improve their expected score by reporting anything other than their true belief. That makes scores comparable and honest.

Formally. The Brier score is $\frac{1}{n} \sum_t (\hat{p}_t - y_t)^2$ and the log loss $-\frac{1}{n} \sum_t [y_t \ln \hat{p}_t + (1 - y_t) \ln(1 - \hat{p}_t)]$. Both are strictly proper. For an event with true probability p, the expected Brier score of a forecast q is $p(1-q)^2 + (1-p)q^2$. The minimum is at $q = p$. Skill against a reference forecast is $1 - \mathrm{BS}/\mathrm{BS}_{\mathrm{ref}}$. Murphy's decomposition writes the Brier score as reliability minus resolution plus uncertainty (§7.2).

Why it appears here. Every Brier skill score in §5–§7 is this quantity, with climatology or the stale label as the reference.

Deeper. Gneiting & Raftery (2007); Murphy (1973).

A.6 ROC curves and AUC

The idea. A classifier's score becomes a yes-or-no decision at any threshold. The receiver operating characteristic (ROC) curve shows, for every threshold, the share of positives caught against the share of negatives falsely flagged. The area under it summarises ranking quality in one number.

Formally. The true-positive rate is $\mathrm{TPR}(c) = \Pr(\hat{p} > c \mid y = 1)$ and the false-positive rate is $\mathrm{FPR}(c) = \Pr(\hat{p} > c \mid y = 0)$. The ROC curve plots TPR against FPR as the threshold c varies. The area under it satisfies $\mathrm{AUC} = \Pr(\hat{p}_i > \hat{p}_j \mid y_i = 1, y_j = 0)$, with ties counted as one half. This is the Mann–Whitney statistic. Any increasing transformation of the scores leaves the AUC unchanged. Suppose the scores of positives and negatives are normal with equal variance and means d' standard deviations apart. Then $\mathrm{AUC} = \Phi(d'/\sqrt{2})$, where Φ is the standard normal distribution function. The precision–recall curve plots $\Pr(y = 1 \mid \hat{p} > c)$ against $\mathrm{TPR}(c)$. Unlike the ROC curve, it depends on the base rate.

Why it appears here. AUC is the ranking score of §1.4, §6.4 and §7.3. It ignores calibration, so the chapter pairs it with the Brier score.

Deeper. Fawcett (2006); Saito & Rehmsmeier (2015).

A.7 Cohen's kappa

The idea. Two labels can agree on most days simply because both say "calm" most of the time. Kappa corrects for that, measuring agreement beyond what the two base rates would produce by chance.

Formally. Let p_o be the observed share of days on which two binary labels agree. Let $p_e = \pi_1 \pi_2 + (1 - \pi_1)(1 - \pi_2)$ be the agreement expected if they were independent with base rates π_1 and π_2. Then $\kappa = (p_o - p_e)/(1 - p_e)$. Kappa is

1 for perfect agreement and 0 for chance-level agreement, and it can be negative. Its maximum attainable value falls when the two base rates differ.

Why it appears here. The agreement table of §6.2 and the stability check of §4.9.

Deeper. Cohen (1960).

A.8 Information coefficient and hit rate

The idea. Judge a forecast of returns by how well it ranks or correlates with what happens. Even a good forecast gets the sign right only slightly more than half the time.

Formally. The information coefficient is the correlation between forecast and outcome. If both are jointly normal with correlation IC, the hit rate is the probability that they share a sign. It equals $\frac{1}{2} + \arcsin(\mathrm{IC})/\pi$, which is 51.6% at an IC of 0.05.

Why it appears here. It sets the scale of what a direction label can offer (§1.4).

Deeper. Section 2 of Edge as Information.

A.9 Sharpe ratio and drawdown

The idea. The Sharpe ratio measures return per unit of risk. The drawdown measures the worst fall from a peak. The first describes the average outcome and the second the worst.

Formally. The annualised Sharpe ratio of daily excess returns e_t is $\sqrt{A}\,\bar{e}/s_e$, with $\bar{e}$ and s_e their mean and standard deviation. The drawdown at t is $1 - W_t/\max_{s \le t} W_s$ for a wealth path W, and the maximum drawdown is its largest value over the sample.

Why it appears here. The value of the bull–bear label, known in advance or when confirmed, is measured by both (§1.1, §6.3).

Deeper. Section 4 of Edge as Information; Appendix A.25 of the companion chapter, on the Sharpe ratio's standard error.

A.10 Logistic regression

The idea. Model the log-odds of an event as a weighted sum of inputs. The model produces probabilities between 0 and 1, and its coefficients show how much each input moves the odds.

Formally. $\Pr(y_t = 1 \mid x_t) = \Lambda(\beta_0 + \beta^\top x_t)$ with $\Lambda(u) = 1/(1+e^{-u})$, so $\ln[p/(1-p)] = \beta_0 + \beta^\top x_t$. The coefficients maximise the log-likelihood $\sum_t [y_t \ln \hat{p}_t + (1-y_t)\ln(1-\hat{p}_t)]$. This function is concave, so Newton's method (iteratively reweighted least squares) finds the unique maximum in a few steps. The probit model replaces Λ with Φ. The ordered logit applies one coefficient vector to $K-1$ cumulative splits of ordered classes. Dynamic versions add lagged labels or a lagged index as inputs.

Why it appears here. It is the default classifier of §5.4 and the model of the worked example (§6).

Deeper. Hastie, Tibshirani & Friedman (2009), section 4.4.

A.11 Shrinkage

The idea. With few independent observations, estimated coefficients are noisy, and a model that uses them as estimated fits noise. Shrinkage moves the coefficients toward zero, trading a little bias for much less variance.

Formally. Ridge regularisation adds $\frac{1}{2}\,\alpha\, n \|\beta\|^2$ to the negative log-likelihood of n rows (§5.4); the lasso uses a multiple of $\|\beta\|_1$ instead and sets some coefficients exactly to zero. As the penalty grows, the fitted probabilities move toward the base rate. Validation chooses the penalty or, as in the worked example, it is fixed in advance.

Why it appears here. In the worked example the twelve-input model lost to a one-input model without shrinkage and matched it with shrinkage (§5.4, §6.4). The reason is that the effective sample is a few dozen episodes.

Deeper. Hastie, Tibshirani & Friedman (2009), section 3.4; Appendix A.31 of the companion chapter, on shrinkage as partial pooling.

A.12 Calibration

The idea. A score that ranks days well need not be a probability. Calibration maps scores to probabilities that match observed frequencies.

Formally. Platt scaling fits $\Pr(y = 1 \mid s) = \Lambda(a + b\,s)$ to a score s by logistic regression. Isotonic regression fits the best non-decreasing step function of the score. Fit both on data the scoring model did not train on, either out-of-fold or on a later period. A flexible model's scores on its own training rows are overconfident.

Why it appears here. Post-processing (§5.11) and the reliability check of §7.2. Tree ensembles need it most (§5.5).

Deeper. Niculescu-Mizil & Caruana (2005).

A.13 Overlapping labels

The idea. A label defined over the next 63 days shares 62 days of outcome with the label of the following day. A training set of thousands of daily rows then holds far fewer independent pieces of evidence than it appears to.

Formally. For labels over windows of length h, rows at least h days apart share no outcome days, so n days contain roughly n/h non-overlapping labels. When the label is also persistent beyond its window, the effective count is lower still. For a rare regime it approaches the number of episodes. López de Prado's *average uniqueness* weights each row by the inverse of the number of labels its window overlaps.

Why it appears here. It explains why §4.10 purges, weights and sizes test sets by episodes. It also explains why the worked example's intervals resample years (§6.1).

Deeper. López de Prado (2018), chapter 4; Appendix A.30 of the companion chapter, on effective sample size.

A.14 Censoring and provisional labels

The idea. At the end of any sample, some labels are not yet decided. Either the forward window has not elapsed, or the current phase has not ended. Treating the undecided labels as decided is a look-ahead in reverse. It assumes an outcome that has not happened.

Formally. A label with fixed reach b is *censored* for the last b days of the sample. A turning-point label is final up to the current running extreme and provisional after it (§4.10). Survival analysis treats the same problem for durations. A spell still in progress has a known lower bound on its length but no end.

Why it appears here. §4.1, §4.4 and the training rule of §4.10.

Deeper. Any survival-analysis text on right-censoring.

A.15 Purging, embargo and walk-forward validation

The idea. In time-series validation, information leaks from test to training through overlapping labels and through serial dependence. Purging removes the leaking rows. An embargo adds a gap. Walk-forward testing predicts the future from the past only.

Formally. Walk-forward validation refits at dates $\tau_1 < \tau_2 < \ldots$ on rows before each τ_k and tests on $[\tau_k, \tau_{k+1})$. Purging drops any training row whose label window overlaps the test period. An embargo also drops training rows within a fixed gap after it. For a label with reach b, the purge at a refit date τ removes training rows with $t > \tau - b$.

Why it appears here. §4.10 and the worked example's training rule ("rows whose label was final by the refit date").

Deeper. López de Prado (2018), chapters 7 and 12; section 11 of the companion chapter and its Appendix A.39.

A.16 Block bootstrap and permutation nulls

The idea. The ordinary bootstrap resamples days independently. This destroys the persistence that makes neighbouring days alike. Resampling blocks of consecutive days keeps the persistence and gives valid error bars for dependent data. A permutation null shows what score the same pipeline would get if features and labels were unrelated but each kept its own persistence.

Formally. The moving-block bootstrap draws blocks of fixed length with replacement and concatenates them. The stationary bootstrap draws blocks of random, geometrically distributed length (Politis & Romano, 1994). A persistence-preserving

permutation shifts the label series circularly against the features by a random offset, refits and records the score. Repeated, it gives the score's distribution under no relationship.

Why it appears here. The 90% intervals of §6.4 resample one-year blocks. §7.6 recommends both tools.

Deeper. Politis & Romano (1994); Appendix A.41 of the companion chapter.

A.17 Multiple testing

The idea. Try enough label definitions, thresholds and inputs and one of them will look good by luck. The best of many trials overstates what any one of them will deliver.

Formally. If a researcher tests N independent strategies with no edge, the expected best Sharpe ratio grows roughly with $\sqrt{2 \ln N}$ times its standard error. Corrections either raise the significance threshold with the number of trials or deflate the best result for them.

Why it appears here. §3.1's warning against testing forty inputs on thirty episodes, and the trial count of §7.6.

Deeper. Harvey, Liu & Zhu (2016); Bailey & López de Prado (2014); Appendix A.40 of the companion chapter.

A.18 Principal components and factor models

The idea. When many inputs move together, a few combinations of them carry most of the variation. Principal components find those combinations. A factor model treats them as unobserved drivers. The Mahalanobis distance measures how unusual a day is after accounting for the inputs' co-movement.

Formally. For standardised inputs with correlation matrix C, the principal components are the eigenvectors of C. The share of total variance explained by the k-th is its eigenvalue divided by the sum of all eigenvalues. A dynamic factor model writes each input as loadings times a few latent factors that follow their own time-series process. The squared Mahalanobis distance of a vector R from mean $\bar{R}$ with covariance Σ is $(R - \bar{R})^\top \Sigma^{-1} (R - \bar{R})$.

Why it appears here. The STLFSI4 is a first principal component, and the NFCI comes from a dynamic factor model (§2.7). Turbulence is a Mahalanobis distance, and the absorption ratio is a principal-component share (§2.7, §5.3). §3.13 measures the input space with them.

Deeper. Jolliffe (2002); Appendix A.23 of the companion chapter.

A.19 Clustering

The idea. Clustering groups days whose features look alike, without labels. Treat each group as a regime.

Formally. k-means chooses K centroids and assigns each day to the nearest, minimising the total squared distance. A Gaussian mixture gives each cluster its own mean and covariance and assigns days probabilistically, fitted by the EM algorithm. Clustering whole return distributions instead of feature vectors needs a distance between distributions, such as the Wasserstein distance (Horvath, Issa & Muguruza, 2024). None of these methods uses the time order of days, so their regimes flicker unless persistence is added (A.26).

Why it appears here. Model-derived labels (§4.6) and the Gaussian-mixture regimes of §2.8.

Deeper. Hastie, Tibshirani & Friedman (2009), sections 8.5 and 14.3; Appendix A.22 of the companion chapter.

A.20 Trees and boosting

The idea. A decision tree splits the input space by thresholds and predicts a constant in each region. Combining many trees, by averaging or by adding them in sequence, gives a flexible model of interactions and non-linear effects.

Formally. A random forest averages deep trees grown on bootstrap samples with random subsets of inputs (Breiman, 2001). Gradient boosting adds shallow trees one at a time, each fitted to the gradient of the loss at the current prediction (Friedman, 2001). Because each tree predicts a constant beyond the last split, the ensemble cannot extrapolate past the range of its training data.

Why it appears here. §5.5, and the student model of the teacher–student pipelines in §5.8.

Deeper. Hastie, Tibshirani & Friedman (2009), chapters 10 and 15; Appendix A.33 of the companion chapter.

A.21 Quantile regression

The idea. Ordinary regression models the average outcome. Quantile regression models a chosen quantile, such as the worst 5% of outcomes, so it can describe how the lower tail moves with conditions even when the average does not.

Formally. The q-th quantile regression minimises $\sum_t \rho_q(y_t - x_t^\top \beta)$ with the check loss $\rho_q(u) = u\,(q - \mathbb{1}[u < 0])$, which weights positive errors by q and negative errors by $1 - q$.

Why it appears here. Growth-at-risk (§2.7) and the direct forecast of drawdown risk in §5.10.

Deeper. Koenker & Bassett (1978).

A.22 Hysteresis

The idea. A switch with a single threshold flips back and forth when its input stays near the threshold. Hysteresis uses two thresholds: a higher one to switch on and a lower one to switch off.

Formally. With entry threshold u_{in} above exit threshold u_{out}, the state turns on when the input reaches u_{in} and turns off when it falls below u_{out}. Otherwise it stays as it was (§5.2). The gap $u_{\text{in}} - u_{\text{out}}$ trades delay against the number of switches.

Why it appears here. The real-time ±20% rule is a hysteresis rule (§1.4), and hysteresis is the cheapest way to reduce flickering regimes (§4.2, §5.11).

Deeper. Section 12 of the companion chapter, on hysteresis bands and trading costs.

A.23 Two-sided filters

The idea. Many smoothing methods estimate the trend at a date from data on both sides of it. They look clean in history but cannot be computed in real time, and their recent values change as new data arrive.

Formally. A two-sided filter computes $\tilde{x}_t = \sum_{k=-m}^{m} w_k x_{t+k}$, using m future values. At the end of a sample it must use a one-sided version, so its last values change as data accumulate. The Hodrick–Prescott filter chooses the trend τ_t to minimise $\sum_t (x_t - \tau_t)^2 + \lambda_{\text{HP}} \sum_t (\Delta^2 \tau_t)^2$ over the whole sample, so every fitted value depends on every observation.

Why it appears here. Labels defined by two-sided filters share the look-ahead of turning-point labels (§4.4), and the OECD's leading indicators use Hodrick–Prescott detrending (§2.8).

Deeper. Hamilton (2018).

A.24 Markov chains and durations

The idea. A Markov chain moves between a few states with fixed probabilities that depend only on the current state. The probability of staying put sets how long states last.

Formally. With transition probabilities $p_{jk} = \Pr(s_{t+1} = k \mid s_t = j)$, a visit to state j lasts a geometric number of days with mean $1/(1 - p_{jj})$. A daily stay probability of 0.98 gives visits averaging 50 days. The stationary distribution gives the long-run share of time in each state.

Why it appears here. The Markov filter that smooths a classifier's output (§5.11) and the latent-state models of §5.7.

Deeper. Appendices A.4 and A.6 of the companion chapter.

A.25 Hidden Markov models and filtering

The idea. A hidden Markov model treats the regime as an unobserved Markov state and infers it from data. Inference comes in three forms that use different information: *predicted* (data up to yesterday), *filtered* (up to today) and *smoothed* (the whole sample).

Formally. The filter alternates two steps. First it carries yesterday's belief through the transition matrix. Then it reweights each state by how well the state explains today's observation and renormalises (the Hamilton filter). The smoother adds a backward pass that uses later data (the Kim smoother). Dividing a classifier's probability by its training prior gives a quantity proportional to the likelihood of the observation in each state. This quantity can replace the reweighting step (§5.11).

Why it appears here. The dating, nowcasting and forecasting tasks of §1.5 correspond to the smoothed, filtered and predicted beliefs. Smoothed probabilities used as labels (§4.6) carry future data.

Deeper. Sections 5 and 6 and Appendices A.10–A.11 of the companion chapter; Bourlard & Morgan (1994) for the hybrid with a classifier.

A.26 Statistical jump models

The idea. A jump model assigns each day to the state whose typical features it is closest to, as clustering does, but charges a fixed penalty for every switch, so states persist.

Formally. It chooses states $s_1, \dots, s_T$ and centroids $\mu_1, \dots, \mu_K$ to minimise $\sum_t \|x_t - \mu_{s_t}\|^2 + \lambda \sum_t \mathbb{1}[s_t \neq s_{t-1}]$. Given the centroids, the best state sequence is found exactly by dynamic programming. It uses the same recursion as the Viterbi algorithm for hidden Markov models. Given the states, each centroid is the mean of its days. Running the recursion forward only, and reading off the best state at each date, gives the causal version.

Why it appears here. Model-derived labels (§4.6), teacher–student pipelines (§5.8) and the worked example's jump-model label (§6).

Deeper. Bemporad et al. (2018); Nystrup, Lindström & Madsen (2020); section 5 of the companion chapter.

A.27 Change-point detection

The idea. Instead of asking which of a few known states the market is in, ask whether its behaviour has just changed.

Formally. A cumulative-sum (CUSUM) detector adds up evidence for a shift with $C_t = \max(0, C_{t-1} + \ell_t)$, where ℓ_t is a log-likelihood ratio of "changed" against "unchanged". It raises an alarm when C_t crosses a threshold (Page, 1954). Bayesian online change-point detection keeps a posterior over the run length, the time since the last change, and updates it with each observation (Adams & MacKay, 2007).

Why it appears here. §5.9, as a monitoring layer beside a classifier.

Deeper. Appendix A.21 of the companion chapter.

A.28 Knowledge distillation

The idea. A large or awkward model, the teacher, can train a simpler one, the student, by letting the student imitate the teacher's outputs. Imitating the teacher's probabilities passes on more than imitating its hard decisions, because the probabilities carry the teacher's uncertainty.

Formally. The student minimises a loss against the teacher's soft targets, for example the cross-entropy $-\sum_t \sum_k q_t^{(k)} \ln \widehat{p}_t^{(k)}$ with $q_t^{(k)}$ the teacher's probability of class k, instead of against one-hot labels.

Why it appears here. Teacher–student regime pipelines (§5.8) are distillation with a two-sided teacher, and soft labels (§4.10) are its targets.

Deeper. Hinton, Vinyals & Dean (2015).

A.29 Realised volatility

The idea. Volatility cannot be observed directly. The sum of squared returns over a window estimates it, and finer sampling of the window improves the estimate.

Formally. $\mathrm{RV}_t^{(n)} = \sqrt{(A/n) \sum_{i=0}^{n-1} \ell_{t-i}^2}$ annualises the average squared daily log return over n days. With intraday data, summing squared five-minute returns gives a far more precise daily variance. The heterogeneous autoregressive (HAR) model forecasts realised variance from its daily, weekly and monthly averages.

Why it appears here. It is the main price-path input (§3.2), the stale label for volatility targets (§1.4), and the basis of the forward-volatility label (§4.5).

Deeper. Andersen, Bollerslev, Diebold & Labys (2003); Corsi (2009); Appendix A.18 of the companion chapter.

A.30 Implied volatility, the VIX and the variance risk premium

The idea. Option prices imply how much volatility buyers expect and how much they will pay for protection against it. The VIX computes this from S&P 500 options for the next 30 days. Option buyers pay for insurance, so implied variance is usually above the variance then realised. The gap is the variance risk premium.

Formally. $\text{VIX}_t^2/10^4$ approximates the expectation of the annualised variance of the S&P 500 over the next 30 days. The expectation uses *risk-neutral* probabilities, which are the probabilities implied by option prices. The calculation uses a strip of out-of-the-money options. The variance risk premium is that expectation minus the expectation under real-world probabilities. A common proxy subtracts trailing realised variance. The VIX *term structure* compares the same measure over 9 days, 30 days, 3 months and 6 months. It normally slopes upward (contango). In stress it inverts (backwardation).

Why it appears here. The VIX is the leading forward-looking input (§3.3), and the volatility regimes of §2.3 are bands on it.

Deeper. Sections 5 and 6 of Implied Volatility; the Cboe VIX methodology; Bekaert & Hoerova (2014).

A.31 Credit spreads and the excess bond premium

The idea. A corporate bond yields more than a government bond of similar maturity for three reasons. It might default. Investors demand a premium for bearing that risk. It is harder to trade. The extra yield, the spread, rises with all three.

Formally. A yield spread subtracts a government yield from a corporate yield. The Baa–10-year spread uses Moody's Baa index and the 10-year Treasury. An option-adjusted spread is the constant spread over the government curve that prices a bond. It removes the value of options embedded in the bond, such as the issuer's right to call it. The excess bond premium is the part of corporate spreads left after accounting for each firm's expected default, so it isolates the price of risk.

Why it appears here. Credit is the price-of-risk input of §3.4 and the second factor of §3.13.

Deeper. Sections 8 and 9 of Bonds and Bond Markets; Gilchrist & Zakrajšek (2012).

A.32 The term spread

The idea. Long-term interest rates embed expected future short rates. When markets expect the central bank to cut, typically because they expect a slowdown, long rates fall below short rates and the curve inverts.

Formally. The term spread is a long yield minus a short yield, for example the 10-year minus 3-month Treasury yield. The near-term forward spread compares the forward rate implied for six quarters ahead with the current short rate. Both also contain a term premium, the extra return demanded for holding long bonds, which varies over time.

Why it appears here. It is the classic recession predictor (§2.8, §3.5) and the slow second factor of §3.13.

Deeper. Section 11 of Bonds and Bond Markets; Estrella & Mishkin (1998).

A.33 Stress and conditions indices

The idea. Financial stress appears at once in many places. Spreads widen, volatility rises and funding markets strain. A stress index averages many such measurements so that the common movement is easier to separate from the noise in each.

Formally. Each input is standardised against its history. The index is a weighted sum, $I_t = \sum_j w_j \tilde{x}_{j,t}$, with weights from a first principal component, a dynamic factor model, or a correlation-based scheme (§5.3). Indices are usually scaled so that zero is the historical average.

Why it appears here. §2.7 uses them as regimes and §3.6 as inputs. Their revisions and full-sample weights are the main hazards.

Deeper. Kliesen & Smith (2010); Brave & Butters (2012); Hollo, Kremer & Lo Duca (2012); Monin (2019).

A.34 Survivorship bias

The idea. A data set that contains only the survivors, such as today's index members or funds still in business, describes a past better than the one that happened. The data set dropped the losers.

Formally. If membership at the end of the sample depends on performance during it, any statistic over today's members conditions on an outcome that was not known at the time. This is a form of look-ahead.

Why it appears here. Breadth and correlation inputs built from current index members (§3.7, §3.8).

Deeper. Brown, Goetzmann, Ibbotson & Ross (1992).

A.35 Variance ratio and Hurst exponent

The idea. If returns are independent, the variance of q-day returns is q times the variance of one-day returns. Departures from that scaling reveal trending or mean reversion.

Formally. $\mathrm{VR}(q) = \mathrm{Var}(r_{t-q+1} + \cdots + r_t)/[q\,\mathrm{Var}(r_t)]$, which equals $1 + 2\sum_{k=1}^{q-1}(1 - k/q)\rho_k$ in terms of the return autocorrelations ρ_k. The ratio is above one when they are positive (trending) and below one when they are negative (mean reverting). The Hurst exponent H describes the same scaling as a power law, with the range of cumulative deviations growing like n^H. For independent returns, $H = \frac{1}{2}$.

Why it appears here. The trending-versus-mean-reverting regimes of §2.5.

Deeper. Lo & MacKinlay (1988).

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